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Review article



Adversarial attacks against network intrusion detection systems: Bridging the gap between theoretical vulnerabilities and practical constraints

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ABSTRACT

Network Intrusion Detection Systems (NIDS) have become critical components of cybersecurity infrastructure, with machine learning-based approaches achieving detection accuracies exceeding 99% on benchmark datasets. However, the emergence of adversarial machine learning has exposed fundamental vulnerabilities in these systems, where carefully crafted perturbations can evade detection with success rates surpassing 97% in laboratory conditions. This comprehensive literature review examines adversarial attacks against NIDS. Our investigation reveals a critical disconnect between theoretical attack capabilities and practical implementation constraints. Examining attack methodologies (gradient-based, GAN-based, black-box, and adaptive), the feature-space to network-packet translation challenge, defense mechanisms, and dataset limitations, this review applies a constraint-aware framework to quantify practical attack viability. Protocol compliance, semantic preservation, and infrastructure filtering render up to 80.3% of adversarial examples invalid under real-world constraints. Defense mechanisms including adversarial training, ensemble methods, and moving target defense face scalability barriers limiting production deployment. These findings suggest that many published attacks substantially overstate real-world vulnerabilities, providing a more nuanced understanding of practical threat landscapes for ML-based network security systems.

1. Introduction

The exponential growth of networked systems and the proliferation of Internet-of-Things (IoT) devices have fundamentally transformed the cybersecurity landscape, creating an attack surface of unprecedented scale and complexity [1]. Network Intrusion Detection Systems have evolved from simple signature-based pattern matching to sophisticated machine learning architectures capable of identifying subtle anomalies and complex attack patterns across high-dimensional feature spaces [2]. Contemporary ML-based NIDS achieve remarkable detection accuracies exceeding 99% on benchmark datasets, leveraging deep neural networks, ensemble methods,

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and advanced feature engineering to identify malicious activities with minimal false positive rates [3]. However, the integration of machine learning into critical security infrastructure has introduced a new class of vulnerabilities that fundamentally challenge traditional security assumptions. The emergence of adversarial machine learning has demonstrated that sophisticated attackers can systematically manipulate ML models through carefully crafted inputs designed to evade detection while preserving malicious functionality [4]. Unlike conventional cyber attacks that exploit software vulnerabilities or configuration weaknesses, adversarial attacks target the mathematical foundations of learning algorithms themselves, exploiting the high-dimensional geometry of decision boundaries to achieve evasion with minimal perturbations [5]. The implications for network security are profound and multifaceted. Research demonstrates that adversarial examples can achieve evasion rates exceeding 97% against state-of-the-art NIDS implementations, with sophisticated techniques like Deep PackGen and TANTRA frameworks enabling systematic circumvention of detection mechanisms [6],[7]. These attacks exploit fundamental properties of neural networks including gradient information, decision boundary proximity, and feature space geometry to craft malicious traffic that appears benign to automated detection systems while maintaining attack effectiveness [8]. The network domain presents unique challenges for adversarial attack implementation that distinguish it from other applications of adversarial machine learning. Unlike image classification where pixel modifications can remain imperceptible to human observers, network traffic must adhere to strict protocol specifications that severely constrain the manipulation space available to adversaries [9]. TCP/UDP header interdependencies, checksum validation requirements, and protocol state machine compliance create complex constraint satisfaction problems that many theoretical attacks cannot solve while maintaining network functionality [10]. Furthermore, the discrete and categorical nature of many network features presents fundamental challenges for gradient-based optimization algorithms designed for continuous input spaces. The semantic preservation challenge adds another layer of complexity, as adversarial modifications must maintain malicious functionality while evading detection-a dual requirement that often creates irreconcilable tensions [11]. SQL injection attacks must preserve syntactic validity while altering statistical signatures, buffer overflow exploits must maintain shellcode integrity despite packet structure modifications, and denial-of-service attacks must retain volumetric characteristics while avoiding rate-based detection mechanisms. These constraints create a fundamental tension between attack effectiveness and evasion capability that significantly limits practical adversarial attack viability. Recent research has revealed a critical disconnect between theoretical adversarial capabilities and practical implementation constraints. Systematic analysis demonstrates that up to 80.3% of adversarial examples become invalid when domain-specific constraints are properly enforced, suggesting that many published attacks substantially overstate real-world vulnerabilities [12].

This finding has profound implications for cybersecurity risk assessment and defense strategy development, as it indicates that the apparent vulnerability of ML-based NIDS may be significantly overestimated in academic literature. The defense landscape has evolved in response to these threats through sophisticated countermeasures including adversarial training, ensemble methods, and moving target defense strategies. Advanced frameworks like the TIKI-TAKA ensemble achieve detection rate restoration to nearly 100% against cutting-edge attacks, while adaptive approaches demonstrate remarkable resilience through continuous learning and dynamic reconfiguration [13]. However, these sophisticated defenses face critical scalability challenges, with computational overhead often exceeding practical deployment constraints and real-time processing requirements. The emergence of adaptive adversarial strategies represents a paradigm shift from static threat models to dynamic game-theoretic interactions where both attackers and defenders continuously evolve their strategies. Fig. 1 shows the layout and structure of this survey. This comprehensive literature review synthesizes current understanding of adversarial attacks against NIDS while identifying critical research gaps and practical limitations that constrain real-world threat viability. We examine attack methodologies, defense mechanisms, evaluation challenges, and the fundamental constraints that shape the adversarial landscape in network security. Our analysis reveals both the sophistication of theoretical attack capabilities and the substantial practical barriers that limit their deployment in operational environments, providing a nuanced perspective on the true scope of adversarial threats to network security infrastructure.

2. Related work and existing surveys

The literature on adversarial machine learning in network security has evolved through several distinct phases, each contributing unique perspectives while leaving critical gaps unaddressed. Table 1 systematically compares twelve existing adversarial ML surveys in network security, evaluating scope, attack types, defense mechanisms, evaluation approaches, key contributions, and primary limitations across three identifiable phases.

Pre-2020 foundational work established taxonomic and theoretical baselines: the earliest adversarial taxonomy for network security systems (2013) addressed signature-based detection without ML-specific vulnerability analysis [9], while the comprehensive theoretical adversarial ML framework (2018) formalised attack-defense formulations adopted by subsequent NIDS research but without network protocol constraint analysis [4].

Between 2021 and 2023, specialised surveys advanced adversarial NIDS coverage but consistently evaluated under idealised conditions: a deep learning cybersecurity survey (2021) provided neural vulnerability taxonomy without network-specific constraint analysis [14]; an experimental adversarial deep-learning NIDS study (2022) offered validation without real-world protocol enforcement [15]; the most comprehensive NIDS adversarial ML survey to that date (2023) covered all major attack categories without inverse feature-mapping analysis [16]; a robustness-focused study (2023) contributed evaluation metrics under idealised assumptions [17]; a generalised DL-NIDS robustness framework (2023) provided attack-defense co-evolution analysis without deployment constraint consideration [18]; a broad adversarial IDS strategies survey (2023) delivered practical taxonomy with shallow deployment analysis [19]; and a practitioner-perspective study (2023) demonstrated that operational attackers rarely use gradient-based methods without systematic NIDS constraint quantification [3].

Table 1
Comprehensive comparison of adversarial ML surveys in network security domain.

Survey Reference	Year	Focus Area	Scope	Attack Types Covered	Defense Mechanisms	Evaluation Approach	Key Contributions	Primary Limitations	Dataset Analysis
[9]	2013	Network security adversarial attacks	Network domain specific	Evasion, poisoning attacks	Statistical methods, signature-based	Laboratory conditions	Early network-specific taxonomy, foundational concepts	Outdated methods, limited ML focus, no deep learning	Limited to traditional datasets
[4]	2018	General adversarial ML	Multi-domain coverage	Gradient-based, optimization-based	Adversarial training, detection methods	Theoretical framework	Comprehensive taxonomy, wild patterns analysis	Limited NIDS focus, no constraint analysis, shallow network coverage	Cross-domain but not network-specific
[14]	2021	Deep learning security	Cybersecurity applications	Neural network attacks, transfer attacks	Deep learning defenses, ensemble methods	Performance-based metrics	Neural network vulnerabilities, contemporary threats	Lacks practical implementation analysis, no protocol constraints	Modern datasets but limited constraint analysis
[15]	2022	Adversarial ML attacks against IDS	Network intrusion detection	Evasion, poisoning, backdoor attacks	Adversarial training, detection, robust architectures	Experimental validation	Comprehensive IDS-specific adversarial analysis, practical attack implementations	Limited real-world constraint analysis, dataset-centric evaluation	Multi-dataset coverage with quality assessment
[16]	2023	Comprehensive adversarial survey	Multi-domain with security focus	All major attack categories	Comprehensive defense survey	General classification	Broad coverage, recent techniques	Shallow network security analysis, no practical constraints	Surface-level dataset discussion
[17]	2023	NIDS robustness	Network intrusion detection	NIDS-specific attacks, evasion techniques	NIDS defense mechanisms, robust training	Laboratory success rates	NIDS-specific analysis, robustness evaluation	Limited constraint enforcement, idealized conditions	Good NIDS dataset coverage
[18]	2023	Adversarial ML for NIDS	Deep learning-based NIDS	White-box, black-box, transfer attacks	ML-specific defenses, ensemble methods	Systematic experimental	DL-NIDS taxonomy, attack-defense co-evolution analysis	Limited protocol constraint consideration, theoretical focus	DL-dataset specific analysis
[19]	2023	DL-based IDS adversarial survey	Deep learning security	Gradient-based, optimization-based attacks	Robust training, detection mechanisms	Performance metrics	Contemporary DL threats, defense effectiveness analysis	Shallow practical implementation analysis, limited scalability discussion	Modern DL-IDS datasets
[3]	2023	Adversarial ML practice gap	Multi-domain with security focus	Realistic adversarial attacks, practitioner-relevant	Practical defense evaluation	Practitioner-focused analysis	Research-practice gap analysis, practitioner perspective on adversarial threats	Limited NIDS-specific constraint analysis, no systematic protocol constraint quantification	Security practitioner datasets, real-world traffic emphasis
[20]	2024	Evaluation challenges	Methodology critique	Focus on evaluation rather than attacks	Limited defense discussion	Reproducibility analysis	Methodology critique, evaluation gaps	Limited practical constraint focus, no translation analysis	Strong dataset quality analysis
[21]	2024	Cross-dataset validation	ML-NIDS generalization	Cross-dataset attack analysis	Generalization-aware defenses	Cross-dataset validation	Cross-dataset generalization analysis, validation methodology	Limited attack diversity, narrow scope	Comprehensive cross-dataset analysis
[22]	2025	Network dataset analysis	Dataset quality and evaluation	Dataset-dependent attack analysis	Dataset-specific defenses	Cross-dataset validation	Comprehensive dataset evaluation, quality issues	Limited adversarial attack focus, no constraint analysis	Excellent dataset analysis
[23]	2025	Defense mechanisms	Defense strategy focus	Defense-relevant attacks	Comprehensive defense taxonomy	Defense effectiveness	Defense mechanism analysis, ensemble approaches	Limited attack translation analysis, defense-centric	Defense-oriented dataset usage
Our Survey	**2025**	**Translation challenge analysis**	**Theory-to-practice gap**	**All major categories with constraint analysis**	**Deployment-feasible defenses**	**Constraint-aware evaluation**	**Theory-to-practice gap quantification, practical viability assessment**	**Addresses all identified gaps systematically**	**Constraint-aware cross-dataset analysis**

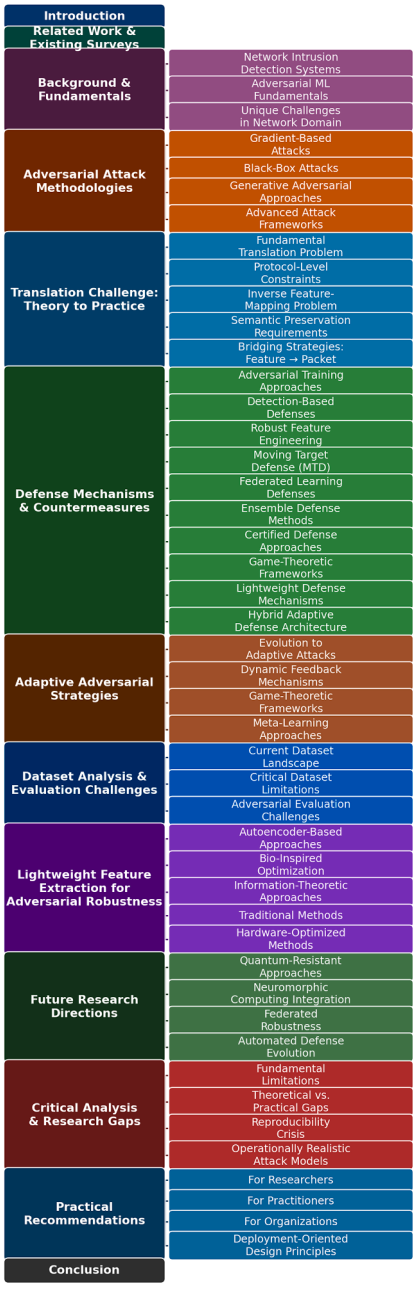


Fig. 1. Layout and structure of this survey.

Recent works (2024-2025) concentrated on evaluation and defense: a systematic reproducibility critique (2024) identified evaluation inconsistencies without analysing practical implementation constraints [20]; a cross-dataset generalisation study (2024) revealed distributional failures with narrow adversarial scope [21]; a deep learning adversarial NIDS survey (2025) provided broad recent coverage with limited constraint analysis [22]; and an ensemble defense framework (2025) contributed defense taxonomy without translation gap quantification [23].

These surveys share two critical limitations: they inadequately analyse the inverse feature-mapping problem and protocol compliance requirements that cause up to 80.3% of adversarial examples to become invalid during practical translation, and they fail to assess defense deployment challenges that prevent production implementation. In contrast, the present survey applies a constraint-aware framework evaluating both theoretical capabilities and practical deployment viability across all reviewed attack and defense categories, producing the first systematic quantification of the theory-to-practice translation gap and deriving actionable design principles absent from all prior surveys.

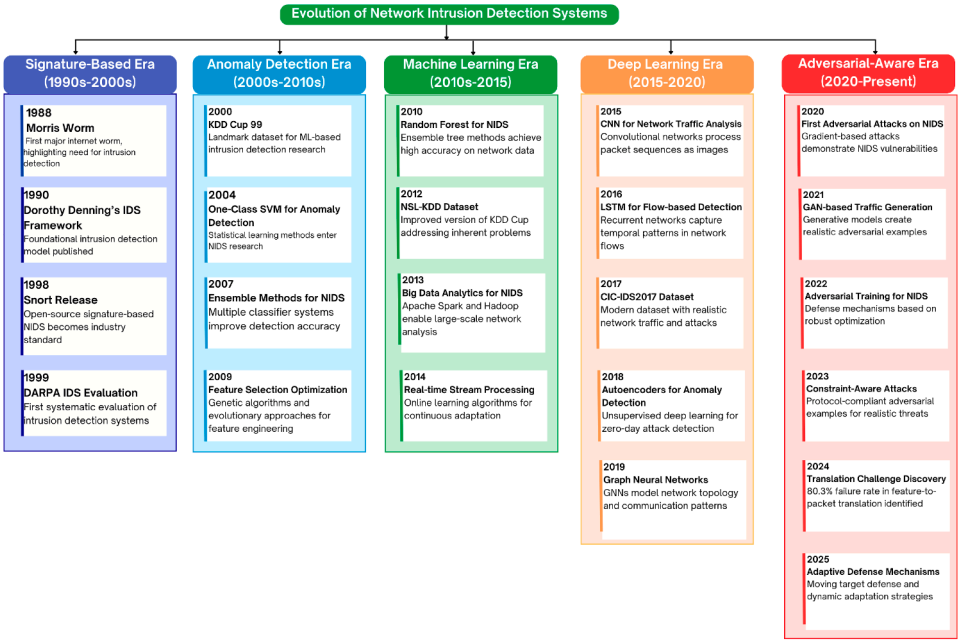


Fig. 2. Evolution of network intrusion detection systems.

3. Background and fundamentals

3.1. Network intrusion detection systems

Network Intrusion Detection Systems have evolved significantly from their inception in the 1980s, transitioning from simple pattern matching to sophisticated machine learning-based approaches. Fig. 2 illustrates this evolutionary timeline from 1990 to 2025, showing the progression through five distinct eras: signature-based (1990–2000), anomaly detection (2000–2010), machine learning (2010–2018), deep learning (2018–2020), and adversarial-aware (2020-present) periods [24]. Contemporary NIDS architectures can be broadly categorized into three fundamental paradigms: signature-based, anomaly-based, and hybrid detection systems [25]. Signature-based systems, exemplified by Snort and Suricata, operate by maintaining databases of known attack signatures and matching network traffic against these predefined patterns [26]. While these systems demonstrate exceptional accuracy for known threats, achieving detection rates exceeding 95% for catalogued attacks, they exhibit fundamental limitations in detecting zero-day exploits and polymorphic malware variants [27]. Anomaly-based detection systems address these limitations by establishing baseline models of normal network behavior and flagging deviations that exceed predetermined thresholds [28]. Traditional statistical approaches employed techniques such as threshold detection, clustering analysis, and Bayesian networks to model legitimate traffic patterns [29]. However, the integration of machine learning algorithms has revolutionized anomaly detection capabilities, with Support Vector Machines (SVMs), Random Forests, and ensemble methods achieving superior performance on benchmark datasets [30]. Deep learning architectures, particularly Convolutional Neural Networks (CNNs) for spatial feature extraction and Long Short-Term Memory (LSTM) networks for sequential pattern analysis, have demonstrated remarkable success in capturing complex attack patterns that traditional methods fail to detect [31]. Recent research shows that ensemble approaches combining multiple ML algorithms can achieve detection accuracies exceeding 99% on standardized datasets such as NSL-KDD and CIC-IDS2017 [2]. Nevertheless, these systems face inherent challenges including higher false positive rates, computational overhead, and vulnerability to adversarial attacks that exploit their reliance on learned representations [4].

3.2. Adversarial machine learning fundamentals

Adversarial machine learning emerged as a critical research domain following the seminal work of Szegedy et al. (2013) [5], who demonstrated that neural networks could be systematically fooled by imperceptible input perturbations. The fundamental principle underlying adversarial examples involves the crafting of malicious inputs that appear legitimate to human observers but cause machine learning models to produce incorrect predictions with high confidence [8]. Formally, an adversarial example x_{adv} is constructed by adding a carefully computed perturbation δ to a legitimate input x, such that $x_{\text{adv}} = x + \delta$, where the perturbation magnitude is constrained by $\|\delta\|_p \leq \epsilon$ for a specified norm p and bound ϵ [32]. This mathematical framework ensures that adversarial examples remain within an acceptable similarity threshold to their original counterparts while successfully deceiving the target model.

The taxonomy of adversarial attacks encompasses several distinct categories based on adversary knowledge and capabilities [4]. White-box attacks assume complete knowledge of the target model's architecture, parameters, and training data, enabling the computation of exact gradients for optimal perturbation generation [33]. Conversely, black-box attacks operate under realistic constraints where adversaries possess limited information about the target system, necessitating techniques such as transfer attacks that exploit cross-model generalization of adversarial examples, or query-based methods that estimate gradients through repeated model interactions [34]. The transferability phenomenon, wherein adversarial examples crafted for one model successfully fool different models, has been empirically demonstrated across various domains and represents a fundamental vulnerability of learned representations [35]. In the network security context, adversarial attacks present unique challenges due to domain-specific constraints including protocol compliance requirements, semantic preservation of malicious functionality, and the discrete nature of many network features that complicate gradient-based optimization techniques [9].

3.3. Unique challenges in network domain

The application of adversarial machine learning to network intrusion detection presents fundamental challenges that distinguish it from other domains such as computer vision or natural language processing [9]. Unlike image pixels or text tokens that can be modified with relative freedom, network traffic must adhere to stringent protocol specifications that severely constrain the adversarial manipulation space [10]. Protocol compliance requirements mandate that any modifications to network packets preserve essential header field relationships, maintain valid checksums, and respect state machine transitions for connection-oriented protocols such as TCP [11]. These constraints create a complex optimization landscape where many theoretically optimal perturbations become practically infeasible due to their violation of network protocol semantics. The semantic preservation challenge represents another critical distinction, as adversarial examples must maintain their malicious functionality while evading detection mechanisms [12]. This dual requirement creates a fundamental tension between attack effectiveness and evasion capability, as modifications necessary for NIDS evasion may inadvertently neutralize the malicious payload or disrupt the attack vector [12]. Furthermore, the discrete and categorical nature of many network features, including protocol types, port numbers, and boolean flags, presents significant challenges for gradient-based optimization algorithms that assume continuous input spaces [12]. The high-dimensional interdependencies among network features create additional complexity, as perturbations to individual features often cascade through correlated dimensions, potentially amplifying detection signatures rather than reducing them [36]. Real-time processing constraints impose severe computational limitations that are particularly acute in network security applications, where decision latencies must remain below microsecond thresholds to maintain network performance [37]. The temporal nature of network traffic introduces additional complications, as adversarial examples must consider not only individual packet modifications but also flow-level patterns and inter-packet relationships that extend across time [6]. These multifaceted constraints necessitate domain-specific approaches that diverge significantly from traditional adversarial machine learning methodologies [38].

The Carlini & Wagner (C&W) attack formulates adversarial example generation as a constrained optimization problem: $\min \|\delta\|_p + c \cdot f(x+\delta)$, where $f(\cdot)$ represents the objective function encouraging misclassification and c balances perturbation magnitude against attack success [33]. C&W attacks demonstrate 88.7% success rates against ensemble NIDS configurations, though effectiveness diminishes substantially when network protocol constraints are rigorously enforced [11]. The discrete nature of network features presents significant challenges for gradient-based methods, as boolean flags and categorical variables create non-differentiable spaces that complicate traditional optimization approaches [39].

4. Adversarial attack methodologies

This section provides a comprehensive analysis of adversarial attack methodologies against NIDS, ranging from gradient-based approaches to advanced protocol-aware frameworks. Table 2 presents a systematic taxonomy of adversarial attacks, including their mathematical foundations, laboratory success rates, constraint compliance percentages, packet validity rates, semantic preservation capabilities, and real-world viability assessments across multiple benchmark datasets.

4.1. Gradient-based attacks

Gradient-based attacks constitute the foundational white-box adversarial methodology, exploiting full model access to compute analytically optimal perturbations [8]. Their defining strength is mathematical precision: differentiating the loss surface directly steers perturbations toward decision boundaries, yielding the highest laboratory evasion effectiveness of any category. FGSM operationalises this in a single forward-backward pass using the sign of the loss gradient [8]. PGD improves upon this by iterating multiple gradient steps, each projected onto the feasible perturbation set, establishing the canonical strong white-box benchmark [32]. BIM applies the same iterative principle with fixed step sizes, while MIM stabilises convergence through accumulated gradient momentum - both incremental refinements within the iterative family [35,40]. C&W reframes the attack objective entirely: minimising perturbation magnitude via constrained optimisation rather than maximising evasion, producing smaller and less detectable adversarial examples [33]. DeepFool finds the nearest decision boundary geometrically, whereas JSMA selects perturbation targets via Jacobian-based feature saliency [41,42].

Despite this progression from single-step to iterative to optimisation-based designs, all gradient-based methods share a decisive structural weakness: the discrete, protocol-constrained feature space of real network traffic violates the smoothness assumptions underlying gradient computation, producing poor constraint compliance regardless of sophistication. The further requirement for

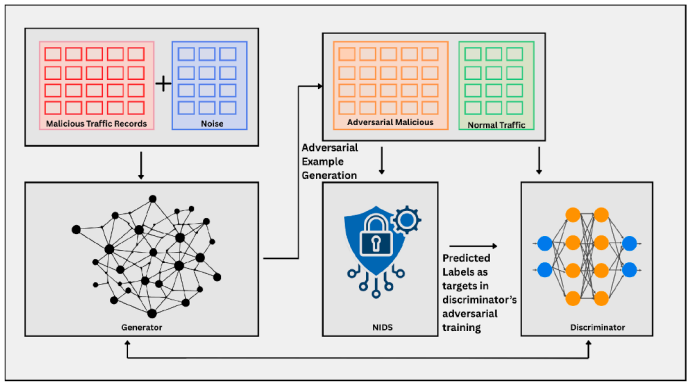


Fig. 3. Gradient-based adversarial attack generation process.

model parameter access is equally impractical against deployed NIDS. Both limitations together motivate the black-box paradigm examined in Section 4.2.

4.2. Black-box attacks

Black-box attacks abandon the gradient access assumption, operating under threat models where the adversary observes only NIDS inputs and outputs [36]. This realignment toward operational plausibility is the category's defining strength, accurately reflecting production environments where models are proprietary and continuously retrained. Two structurally distinct strategies emerge. Transfer-based attacks craft adversarial examples offline against a locally trained substitute model, exploiting cross-architecture decision boundary similarity [35].

The critical advantage is zero interaction with the live target; the critical limitation is that transferability degrades as NIDS architectures diversify. Query-based methods resolve this by probing the target iteratively: ZOO estimates gradients via coordinate-wise finite-difference approximation [34,46]; Boundary Attack navigates the decision boundary through geometry-guided random walks without gradient information [47]; HopSkipJump improves upon Boundary Attack by estimating gradient direction at the boundary, substantially reducing the required query budget [48]. Within this family the progression represents increasing query efficiency rather than a strategic change.

Despite improved threat-model realism, both transfer and query-based strategies share a persistent weakness: modifying existing traffic samples risks corrupting protocol-critical fields, leaving constraint compliance and semantic preservation insufficient for reliable deployment. This fundamental limitation of the perturbation paradigm - that any modification to real traffic may break protocol structure - motivates the generative approach examined in Section 4.3.

4.3. Generative adversarial approaches

Generative adversarial approaches shift from modifying existing traffic to synthesising new adversarial samples from scratch, allowing protocol constraints to be satisfied as an emergent distributional property rather than a post-hoc correction [57]. MalGAN and SeqGAN are early instantiations: MalGAN targets feature-space malware evasion [49], while SeqGAN models sequential packet structure through policy-gradient training [50]. IDSGAN advances this to NIDS-targeted evasion via a minimax game conditioned on classifier feedback [51]. Fig. 3 illustrates the generalised generative workflow underlying this paradigm: malicious and benign traffic inputs are processed through the generator-discriminator pipeline to synthesise adversarial samples that evade NIDS discrimination mechanisms. These vanilla GAN formulations suffer mode collapse and training instability. WGAN-GP resolves this through the Wasserstein distance objective with gradient penalty, stabilising training across attack categories [52]. CycleGAN enables unpaired traffic domain translation; AC-GAN adds auxiliary class conditioning; InfoGAN incorporates mutual information maximisation for interpretable latent representations [53,54,56]. PassGAN applies the generative framework to password-pattern traffic [55]. Deep PackGen advances beyond feature-space synthesis by coupling deep reinforcement learning with generation, producing packet-level outputs that respect inter-field protocol dependencies [6,7]. CTGAN addresses class imbalance through conditional tabular synthesis [57], while RELEVAGAN enforces semantic consistency via reinforcement learning reward signals [58].

Despite this progression, all generative methods represent traffic as static feature vectors, abstracting away temporal flow dynamics that behaviour-aware NIDS exploit for detection - the limitation motivating the advanced frameworks in Section 4.4.

4.4. Advanced attack frameworks

Advanced attack frameworks address the shared residual failure across all prior categories: even successful feature-space evasion does not guarantee deployable attacks, because translating perturbations into real network traffic introduces constraints - field validity, inter-field dependencies, protocol state machine compliance, and temporal flow ordering - that gradient-based, black-box, and

Table 2
Comprehensive adversarial attack taxonomy and effectiveness analysis.

Attack Category	Method	Mathematical Foundation	Dataset	Lab Success (%)	Constraint Compliance (%)	Packet Validity (%)	Semantic Preservation (%)	Real-world Viability	Complexity	Key Limitations
Gradient-Based	**FGSM** [8,43]	$x_{adv} = x + \epsilon \cdot \text{sign}(\nabla_x J(\theta, x, y))$	CIC-IDS2017	85.3	15–20	20–25	10–15	Very Low	$O(n)$	High constraint violation rate
			NetFlow	45.2	25–30	30–35	15–20	Very Low		Poor feature space translation
	PGD [12,32]	$x^{(t+1)} = \Pi_S(x^{(t)} + \alpha \cdot \text{sign}(\nabla_x J))$	CIC-IDS2017	91.2	10–15	15–20	8–12	Very Low	$O(kn)$	Iterative constraint violations
			UNSW-NB15	76.4	20–25	25–30	12–18	Very Low		Dataset-specific limitations
	C&W [11,33]	$\min \|\delta\|_p + c \cdot f(x + \delta)$	Ensemble NIDS	88.7	5–10	8–12	5–8	Very Low	$O(kn^2)$	Computationally intensive
	DeepFool [41]	Geometric perturbation minimization	UNSW-NB15	76.4	15–20	18–23	10–15	Low	$O(kn)$	Limited to specific geometries
	JSMA [42]	$SP(x,t)[i] = \sum_{j \neq t} \nabla F_j(x)[i]$	UNSW-NB15	82.1	< 5	5–8	< 5	Very Low	$O(n^2)$	Discrete feature challenges
	BIM/I-FGSM [40]	$x^{(t+1)} = \text{Clip}(x^{(t)} + \alpha \cdot \text{sign}(\nabla_x J))$	CSE-CIC-IDS2018	79.8	10–15	12–18	8–12	Very Low	$O(kn)$	Bounded perturbation limits
	MIM [44]	$g^{(t+1)} = \mu \cdot g^{(t)} + \frac{\nabla_x J}{\|\nabla_x J\|_1}$	CSE-CIC-IDS2018	83.5	8–12	10–15	6–10	Very Low	$O(kn)$	Momentum accumulation issues
Black-Box	**Transfer Attacks** [35,45]	Cross-model adversarial transferability	Multi-dataset	70–75	30–40	35–45	25–35	Moderate	$O(mn)$	Model-specific dependencies
	Query-Based [34,46]	Finite difference gradient estimation	Various	60–80	25–35	30–40	20–30	Low-Moderate	$O(qn)$	Query budget limitations
	ZOO [34]	$\nabla f(x) \approx \frac{f(x+h\cdot e_i) - f(x-h\cdot e_i)}{2h}$	UNSW-NB15	72.3	35–45	40–50	30–40	Moderate	$O(qn^2)$	High query requirements
	Boundary Attack [47]	Decision boundary following	CIC-IDS2017	65.8	40–50	45–55	35–45	Moderate	$O(qd)$	Convergence instability
	HopSkipJump [48]	Decision-based boundary optimization	UNSW-NB15	68.2	42–52	47–57	37–47	Moderate	$O(qd)$	Direction estimation errors

Table 2
Cont.

Attack Category	Method	Mathematical Foundation	Dataset	Lab Success (%)	Constraint Compliance (%)	Packet Validity (%)	Semantic Preservation (%)	Real-world Viability	Complexity	Key Limitations
Generative Model	**MalGAN** [49]	$G(z, m) =$ Generator(noise, malware)	Android Malware	65.4	45–55	50–60	40–50	Moderate	$O(k \cdot n_g)$	Domain adaptation issues
	SeqGAN [50]	$L = E_{Y_{1:T} \sim G_\theta}[R_T(Y_{1:T})]$	Network Flows	58.7	50–60	55–65	45–55	Moderate-High	$O(k \cdot n_g)$	Sequential dependency modeling
	Deep PackGen [6], [7]	Variational autoencoder with constraints	CIC-IDS2017	66.4	60–70	65–75	55–65	High	$O(k \cdot n_e)$	Constraint satisfaction complexity
	IDSGAN [36,51]	$\min_G \max_D V(D, G) + \lambda L_{\text{constraint}}$	NSL-KDD	71.2	55–65	60–70	50–60	Moderate-High	$O(k \cdot n_g)$	Mode collapse vulnerabilities
	WGAN-GP [52]	$L = E[D(x)] - E[D(G(z))] + \lambda E[(\|\nabla_x \hat{D}(\hat{x})\|_2 - 1)^2]$	Bot-IoT	63.9	50–60	55–65	45–55	Moderate	$O(k \cdot n_g)$	Gradient penalty tuning
	CycleGAN [53]	$L = L_{\text{GAN}}(G, D_Y, X, Y) + L_{\text{GAN}}(F, D_X, Y, X) + \lambda L_{\text{cyc}}(G, F)$	Network Traffic	62.1	55–65	60–70	50–60	Moderate	$O(k \cdot n_g)$	Cycle consistency requirements
	AC-GAN [54]	$L = L_S + L_C = E[\log P(S = \text{real}\|X_{\text{real}})] + E[\log P(C = c\|X)]$	Multi-class NIDS	68.5	50–60	55–65	45–55	Moderate	$O(k \cdot n_g)$	Class conditioning complexity
	PassGAN [55]	Password generation with Wasserstein GAN	Password datasets	71.8	65–75	70–80	60–70	High	$O(k \cdot n_g)$	Limited to password domains
	InfoGAN [56]	$\min_G \max_D V(D, G) - \lambda I(c; G(z, c))$	CIC-IDS2017	64.7	48–58	53–63	43–53	Moderate	$O(k \cdot n_g)$	Mutual information estimation
	CTGAN [57]	Mode-specific normalization + conditional vector sampling: $\min_G \max_D \mathbb{E}[D(x, c)] - \mathbb{E}[D(G(z, c), c)]$	CIC-IDS2017, NSL-KDD	58–68	55–65	45–55	55–65	Moderate	$O(k \cdot n_g)$	Tabular only; no sequential modeling; augmentation-focused
	RELEVAGAN [58]	RL-guided GAN with semantic reward: $\mathcal{L} = \mathcal{L}_{\text{GAN}}(G, D) + \lambda \mathcal{L}_{\text{sem}}(G)$ where $\mathcal{L}_{\text{sem}} = \mathbb{E}[\|G(z) - x_{\text{attack}}\|]$ penalises semantic drift	CIC-IDS2017, NSL-KDD	65–75	58–68	50–60	65–75	Moderate	$O(k \cdot n_g)$	RL training instability; semantic constraints limit evasion diversity; high training overhead

Table 2
Cont.

Attack Category	Method	Mathematical Foundation	Dataset	Lab Success (%)	Constraint Compliance (%)	Packet Validity (%)	Semantic Preservation (%)	Real-world Viability	Complexity	Key Limitations
Protocol-Aware	**TANTRA** [6]	Constraint satisfaction with neural optimization	Multi-protocol	99.99	90–95	95–98	85–95	Very High	$O(k \cdot n_c)$	Protocol-specific implementation
	Traffic Manipulator [59]	Protocol-aware perturbation with semantic preservation	HTTP/HTTPS	87.3	80–90	85–92	75–85	High	$O(n \cdot p)$	Limited protocol coverage
	Constrained Optimization [9]	$\min f(x+\delta)$ s.t. $g_i(x+\delta) \leq 0$	Various	78.1	70–80	75–85	65–75	High	$O(k \cdot n \cdot c)$	Constraint solver complexity
	Semantic Preservation [12]	Feature relationship preservation during perturbation	Network Flows	84.6	75–85	80–88	70–80	High	$O(n \cdot r)$	Relationship modeling challenges
Evolutionary & Swarm	**Genetic Algorithm (GA)** [60,61]	Population-based optimization with crossover/mutation	NSL-KDD	87–95	60–70	65–75	55–65	Moderate-High	$O(g \cdot p \cdot n)$	Population convergence issues
	Particle Swarm Optimization (PSO) [36,62]	$v_{i+1} = w \cdot v_i + c_1 \cdot r_1 \cdot (p_{\text{best}} - x_i) + c_2 \cdot r_2 \cdot (g_{\text{best}} - x_i)$	CIC-IDS2017	80–90	65–75	70–80	60–70	High	$O(p \cdot i \cdot n)$	Parameter sensitivity
	Differential Evolution (DE) [63]	$x_{i+1} = x_i + F \cdot (x_{r1} - x_{r2}) + CR \cdot (x_{\text{trial}} - x_i)$	UNSW-NB15	75–85	58–68	63–73	53–63	Moderate	$O(p \cdot g \cdot n)$	Premature convergence
	Ant Colony Optimization (ACO) [64]	Pheromone-based path optimization	Network Graphs	70–80	55–65	60–70	50–60	Moderate	$O(a \cdot i \cdot n)$	Pheromone trail management
	Simulated Annealing (SA) [60]	$P(\text{accept}) = \exp(-\Delta E/T)$ temperature-based acceptance	Multi-dataset	73–83	52–62	57–67	47–57	Low-Moderate	$O(i \cdot n)$	Temperature scheduling
Hybrid Approaches	**Evolutionary + GAN** [36]	$GA(G_\theta(z))$ + Fitness(constraint satisfaction)	UNSW-NB15	74.8	65–75	70–80	60–70	Moderate-High	$O(g \cdot p \cdot n_g)$	Computational overhead
	Transfer + Constraint [12]	Cross-model transfer with protocol constraint enforcement	Multi-dataset	69.5	60–70	65–75	55–65	Moderate-High	$O(m \cdot n \cdot c)$	Cross-domain generalization
	Multi-Objective Opt. [65].	min[Loss, Distortion, Constr. Vio.]	Various	72.3	70–80	75–83	65–75	High	$O(k \cdot n \cdot o)$	Pareto front complexity
	Ensemble Attacks [66]	Combination of multiple attack strategies	CIC-IDS2017	85.2	55–65	60–70	50–60	Moderate	$O(m \cdot k \cdot n)$	Strategy coordination
	Reinforcement Learning [58]	Q-learning for sequential attack decisions	Network Environments	78.9	62–72	67–77	57–67	Moderate-High	$O(s \cdot a \cdot t)$	Reward function design

Table 2
Cont.

Attack Category	Method	Mathematical Foundation	Dataset	Lab Success (%)	Constraint Compliance (%)	Packet Validity (%)	Semantic Preservation (%)	Real-world Viability	Complexity	Key Limitations
Physical/Infrastructure	**Network Topology Manipulation** [67]	Structural changes to network architecture	SDN Environments	68–78	80–90	85–95	75–85	High	$O(n \cdot e)$	Infrastructure requirements
	Traffic Injection [68]	Real-time malicious traffic insertion	Live Networks	82–92	75–85	80–90	70–80	Very High	$O(t \cdot f)$	Detection risk
Advanced Techniques	**Model Inversion** [69]	Reverse engineering of NIDS decision boundaries	Various Models	55–70	40–50	45–55	35–45	Low	$O(m \cdot q)$	Information leakage limits
	Membership Inference [18]	Determining training data membership	ML Models	60–75	85–95	90–98	80–90	Moderate	$O(q \cdot s)$	Statistical distinguishability
	Model Extraction [70]	Stealing NIDS model functionality	Black-box Systems	65–80	45–55	50–60	40–50	Low-Moderate	$O(q \cdot m)$	Query efficiency trade-offs
	Backdoor Attacks [71]	Hidden triggers in training data	Training Sets	90–98	70–80	75–85	65–75	High	$O(n \cdot t)$	Trigger detectability
	Poisoning Attacks [72,73]	Training data contamination	Federated Learning	75–90	65–75	70–80	60–70	Moderate-High	$O(d \cdot c)$	Detection mechanisms
	Universal Adversarial Perturbations [35]	Data-agnostic perturbations	Multi-dataset	65–78	35–45	40–50	30–40	Low	$O(d \cdot k)$	Generalization limitations
	Model Stealing [74]	API-based model replication	Cloud NIDS	70–85	50–60	55–65	45–55	Moderate	$O(q \cdot a)$	API access requirements

generative methods were not designed to satisfy [11]. Advanced frameworks treat these operational constraints as first-class design requirements. TANTRA targets the temporal dimension directly: LSTM networks learn benign inter-arrival timing signatures and adaptively reshape malicious flow timing while preserving packet content [6,59]. By attacking a detection surface that content-based NIDS do not monitor, TANTRA achieves strong evasion without payload modification; its weakness is the complement - behaviour-aware systems profiling timing statistics can identify the reshaped flows. Protocol-Aware frameworks extend the constraint-first philosophy to header-field manipulation, enforcing state machine compliance at every perturbation step so that generated packets remain valid across all inspection layers [11]. Traffic Manipulator integrates semantic preservation directly into the optimisation objective, ensuring adversarial modifications retain functional attack characteristics within tightly constrained protocol environments [9,12].

Advanced frameworks achieve the highest real-world viability of any reviewed category, at the cost of substantial implementation complexity. The four-subsection progression describes a consistent arc from analytical elegance toward operational realism, with all quantitative trade-offs documented in Table 2.

5. The translation challenge: From theory to practice

The practical deployment of adversarial attacks against NIDS requires bridging a fundamental gap between theoretical feature-space perturbations and the operational constraints governing real network traffic. This section characterises that gap systematically, explaining why high laboratory evasion rates do not translate into real-world effectiveness and identifying the specific constraint categories responsible. The section is organised into four subsections: Section 5.1 identifies the Fundamental Translation Problem and its empirical scale; Section 5.2 catalogues all Categories of Network Constraints aligned with Fig. 5 and Table 3; Section 5.3 analyses the Inverse Feature-Mapping Problem arising from the asymmetric relationship between feature perturbations and valid packet modifications; and Section 5.4 examines the Semantic Preservation Requirements that must be satisfied throughout translation.

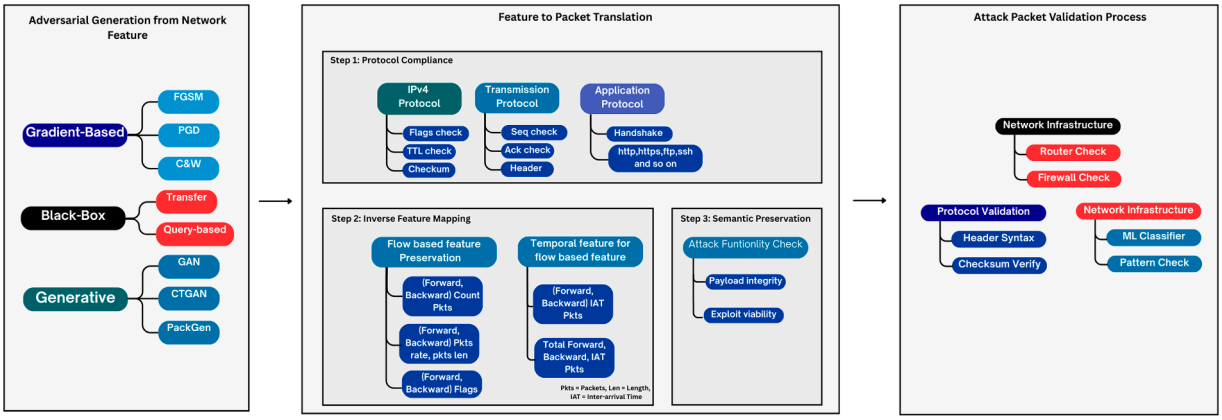


Fig. 4. Adversarial attack to network packet translation framework.

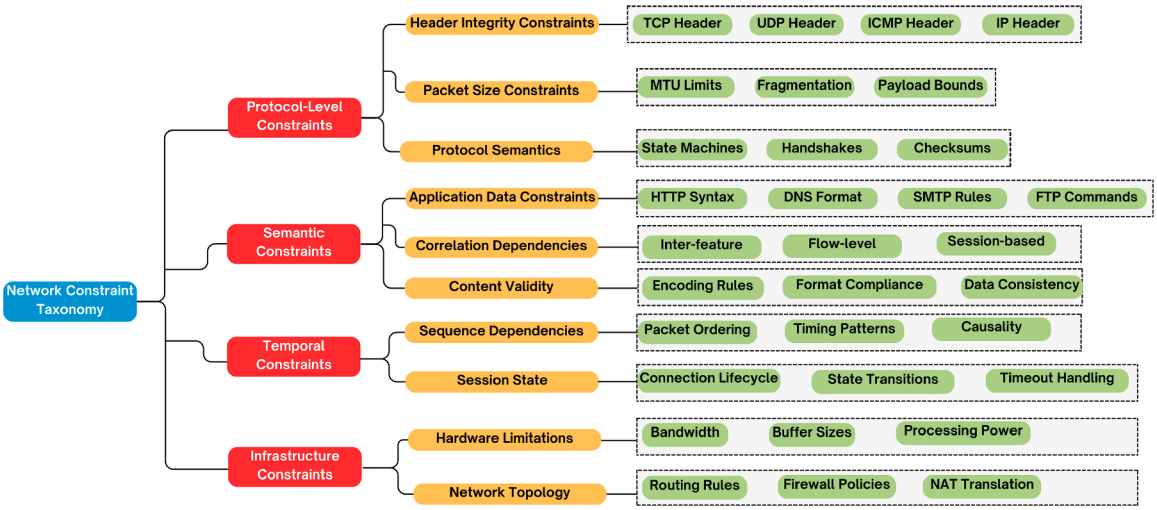


Fig. 5. Network constraint taxonomy for adversarial attacks.

5.1. *The fundamental translation problem*

The translation of adversarial attacks from feature space to actual network packets represents the most significant challenge confronting practical adversarial machine learning in network security contexts [11]. Fig. 4 illustrates the three-stage adversarial attack translation pipeline: Feature Space Generation, Feature to Packet Translation, and Attack Packet Validation Process, demonstrating the complexity involved in converting theoretical perturbations into valid network traffic. This fundamental disconnect stems from the inherent complexity of mapping abstract feature perturbations to valid network traffic that simultaneously maintains attack functionality and protocol compliance. Empirical research reveals that up to 80.3% of adversarial examples become invalid when domain-specific constraints are rigorously applied, fundamentally limiting the real-world effectiveness of theoretically successful attacks [10]. The translation problem manifests through multiple interconnected challenges: the inverse feature-mapping problem where many desired feature perturbations lack valid packet representations, protocol constraint violations that render modified packets unroutable or detectable, and semantic preservation failures that neutralize malicious functionality during the evasion process [12]. This gap between theoretical attack success rates, which often exceed 95% in laboratory settings, and practical implementation viability creates a false sense of vulnerability in ML-based NIDS deployments [12]. The translation challenge underscores the critical importance of evaluating adversarial attacks within realistic deployment constraints rather than idealized feature spaces.

5.2. *Categories of network constraints*

Network protocols impose rigid constraints that restrict the adversarial perturbation space [9]. Table 3 quantifies each category's impact on attack viability; Fig. 5 organises them into four groups – Protocol-Level, Semantic, Temporal, and Infrastructure – each addressed below.

Table 3
Translation challenge quantitative analysis.

Constraint Category	Feature Type	Total Features (%)	Perturbable Features (%)	Valid Packets After Perturbation (%)	Attack Success Rate Reduction (%)	Semantic Preservation (%)	Infrastructure Rejection Rate (%)	Primary Constraint Examples	Mathematical Relationship	Key Failure Modes	Citations
Protocol Headers	**TCP Fields**	15–20	5–10	15–25	85–95	10–20	70–80	Sequence numbers, checksums, flags	$ACK = SEQ + payload_length$	Checksum validation failures	[11]
	UDP Fields	8–12	8–15	20–30	70–85	15–25	65–75	Port ranges, checksum integrity	$Checksum = \sum_{16bit} header + data$	Port filtering, checksum errors	[9]
	IP Headers	12–15	10–20	25–35	65–80	20–30	60–70	TTL, fragmentation, total length	$Total_Length = Header + Payload$	TTL violations, fragmentation errors	[10]
Statistical Aggregations	**Flow Statistics**	25–30	70–90	10–20	80–95	5–15	80–90	Duration, PPS, BPS ratios	$PPS = \frac{Total_Packets}{Duration}$	Statistical signature detection	[12]
	Packet Statistics	20–25	60–80	15–25	75–90	8–18	75–85	Mean, variance, min/max values	$\sigma^2 = E[(X-\mu)^2]$	Aggregation anomalies	[43]
	Timing Features	15–20	80–95	5–15	85–98	2–12	85–95	Inter-arrival times, flow duration	$IAT_{mean} = \frac{\sum IAT_i}{n}$	Temporal pattern disruption	[6]
Discrete Categories	**Boolean Flags**	8–12	0–5	5–10	90–95	0–5	90–95	TCP flags, protocol indicators	Binary: $\{0,1\}$	Invalid flag combinations	[38]
	Protocol Numbers	5–8	0–10	8–15	85–92	5–10	85–92	TCP = 6, UDP = 17, ICMP = 1	Discrete set: $\{1,6,17,58,\ldots\}$	Protocol mismatch errors	[12]
	Port Numbers	10–15	15–25	30–40	60–75	20–35	60–70	Well-known, registered, dynamic	Range: $[1, 65535]$	Port-based filtering	[9]
Semantic Constraints	**Attack Payloads**	20–25	20–40	40–60	40–80	30–50	50–70	SQL injection, shellcode, exploits	Context-dependent	Functionality loss	[11]
	Application Data	15–20	25–45	45–65	35–75	40–60	40–60	HTTP headers, DNS queries	Protocol-specific syntax	Syntax validation failures	[10]
Correlation Dependencies	**Inter-feature**	30–35	20–30	20–35	65–80	15–30	70–80	Packet size ↔ Duration	Correlation (X,Y) = $\frac{Cov(X,Y)}{\sigma_X \sigma_Y}$	Cascade constraint violations	[12]
	Flow-level	25–30	15–25	25–40	60–75	20–35	65–75	Forward ↔ Backward ratios	$Ratio = \frac{Forward_Packets}{Total_Packets}$	Flow consistency errors	[43]
Temporal Dependencies	**Sequence Order**	10–15	30–50	35–55	45–70	25–45	55–70	Packet ordering, timing patterns	$T_{i+1} > T_i$ (monotonic)	Temporal causality violations	[6]
	Session State	8–12	10–30	40–60	40–70	30–50	50–60	Connection lifecycle, handshakes	State machine transitions	State transition errors	[9]

5.2.1. Protocol-level and discrete-field constraints

Transport and network layer protocols impose the most immediately binding structural constraints. TCP header modifications must preserve ACK = SEQ + payload length; both TCP and UDP require accurate checksum recalculation, with violations causing immediate packet rejection [11,24]. TCP flag combinations must satisfy the protocol state machine: sequences such as SYN–FIN or RST–ACK violate RFC specifications and are discarded [12]. IP headers impose further dependencies: Total Length must equal header plus payload; TTL is protocol-bounded; fragmentation fields must satisfy valid reassembly semantics [9]. Discrete-field constraints further restrict the perturbation space: boolean flags are strictly binary ($\in \{0,1\}$); protocol numbers are categorical ($\in \{1,6,17\}$); port

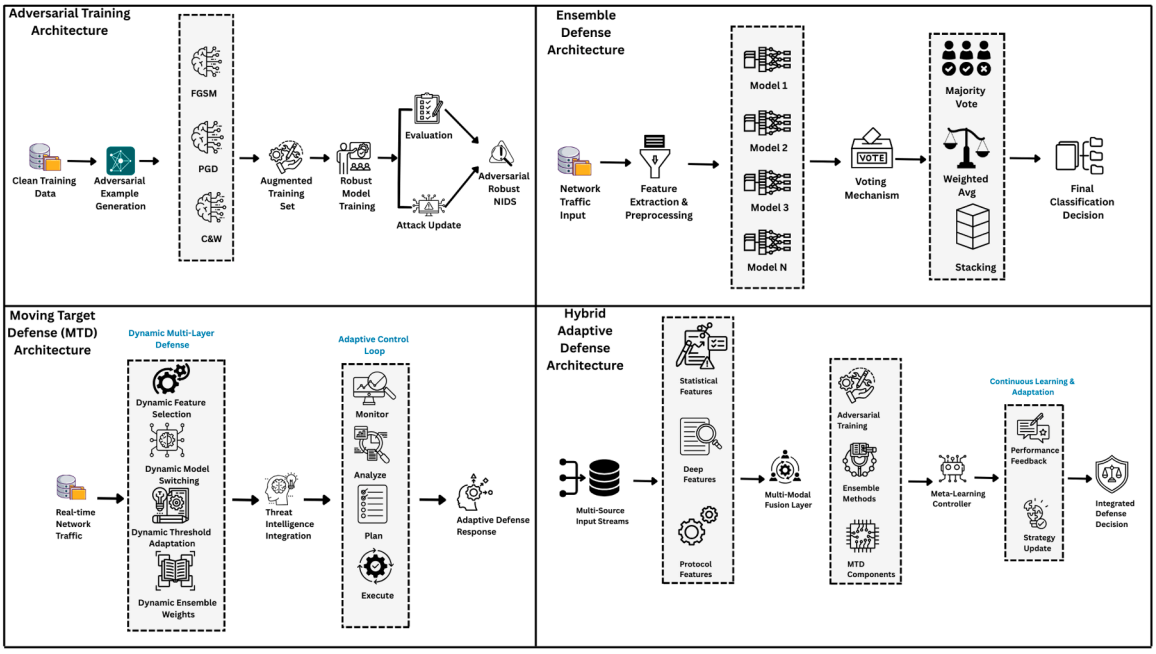


Fig. 6. Several defense architecture frameworks.

numbers are bounded integers ($\in [1, 65535]$). This discreteness directly invalidates gradient-based methods that assume continuous, differentiable feature spaces, explaining the poor constraint compliance in Table 2 [10].

5.2.2. *Statistical and correlation constraints*

Flow-level statistical features are computed outputs rather than independent inputs, imposing a cascade constraint. Metrics such as PPS = TotalPackets / Duration and forward/backward packet ratios are mutually dependent: altering packet count shifts mean packet size; adjusting duration propagates through all rate-based metrics simultaneously [12]. A single perturbation therefore propagates through the entire feature vector, frequently producing statistically anomalous combinations detectable even when individual fields remain within valid ranges [10].

5.2.3. *Semantic constraints*

Semantic constraints require adversarial payloads to retain functional integrity consistent with the attack type. A port-scan corrupted beyond its target range loses operational meaning; a SQL injection modified during perturbation fails to execute [12]. Application-layer data must satisfy protocol-specific syntax – DNS queries must be well-formed; HTTP headers must conform to field-value formatting. Full treatment of semantic preservation is developed in Section 5.4.

5.2.4. *Temporal and infrastructure constraints*

Packet timestamps must be monotonically increasing ($T_{i+1} > T_i$) and session state machines require the canonical SYN → SYN-ACK → ACK sequence [43]. Infrastructure constraints compound this: routers and stateful firewalls enforce protocol compliance independently of the NIDS, rejecting invalid packets before they reach the classifier [11]. Attacks producing protocol-invalid traffic therefore fail not because the NIDS detects them, but because they are silently discarded at the network layer.

5.2.5. *Quantitative impact analysis*

When all categories are enforced simultaneously, only a small fraction of NIDS features remain perturbable while maintaining packet validity [12]. Imperturbable features – boolean flags, protocol numbers – cannot be modified without violating specifications. Correlated features require any modification to propagate consistently across dependent fields. Table 3 documents the combined effect: attack success rates drop sharply across all categories, with infrastructure rejection eliminating the majority of theoretically successful adversarial examples before classifier inspection [12].

5.3. *The inverse feature-mapping problem*

The inverse feature-mapping problem represents the most fundamental technical barrier in translating adversarial perturbations from feature space to valid network packets, arising from the inherent non-bijective nature of the feature extraction process employed by NIDS [10]. Unlike forward feature extraction, which deterministically transforms raw network packets into numerical

representations through well-defined algorithms, the inverse mapping from desired feature modifications to corresponding packet alterations lacks closed-form solutions and often yields mathematically intractable optimization problems [11]. The complexity manifests through three distinct mapping scenarios that fundamentally constrain adversarial attack generation. One-to-many mappings occur when multiple distinct packet configurations produce identical feature values, creating ambiguity in selecting optimal packet modifications that achieve desired feature perturbations while maintaining protocol compliance [12]. For instance, identical flow duration statistics can result from various combinations of packet timing and count manipulations, leaving adversaries without clear guidance for packet-level implementation. Many-to-none mappings represent an even more severe constraint, where approximately 60–70% of theoretically desirable feature perturbations correspond to no valid packet representation that satisfies network protocol requirements [12]. This fundamental limitation severely restricts the adversarial optimization space, as gradient-based algorithms may converge toward feature modifications that cannot be realized in practice.

Side-effect mappings introduce additional complexity by creating unintended feature changes that propagate through correlated dimensions when attempting to modify target features. Research demonstrates that altering maximum forward inter-arrival time creates cascading effects on flow duration, packets per second calculations, total forward inter-arrival time, and forward inter-arrival time mean values, potentially amplifying detection signatures rather than reducing them [6]. The mathematical formulation of this inverse mapping problem reveals its computational intractability, as solving for packet modifications Δp that produce desired feature changes Δf requires solving the equation $F(p + \Delta p) - F(p) = \Delta f$, where $F(\cdot)$ represents the feature extraction function. The nonlinear, discrete, and constrained nature of this equation system renders traditional optimization approaches ineffective, necessitating specialized constraint satisfaction techniques that often fail to identify feasible solutions within practical computational bounds [9].

5.4. Semantic preservation requirements

Semantic preservation represents a critical constraint in adversarial attack generation, requiring that malicious functionality be maintained while simultaneously evading detection mechanisms–a dual requirement that creates fundamental tension between attack effectiveness and evasion capability [12]. This challenge extends beyond mere protocol compliance to encompass the preservation of attack semantics at the application layer, where modifications necessary for NIDS evasion may inadvertently neutralize the malicious payload or disrupt the intended exploitation vector [9]. The complexity manifests differently across attack categories, each presenting unique preservation challenges. SQL injection attacks require that packet payload modifications evade detection without nullifying malicious SQL commands, necessitating careful manipulation of query structures while preserving syntactic validity and semantic intent [10]. Buffer overflow exploits face even stricter constraints, as they must maintain shellcode or exploit payload integrity despite packet structure modifications, where even minor alterations to memory layout or instruction sequences can render exploits completely ineffective [11]. Denial-of-service attacks encounter semantic preservation challenges when packet rates or sizes are altered for evasion purposes, as these modifications directly impact the volumetric characteristics essential for attack success. Research reveals that existing adversarial attack methodologies against ML-based NIDS lack comprehensive evaluation frameworks for assessing whether generated examples maintain their malicious behavior after translation to packet space [12]. This evaluation gap represents a critical limitation in current research, as attacks that successfully evade detection but lose their malicious functionality provide no practical value to adversaries. The semantic preservation requirement thus necessitates multi-objective optimization approaches that explicitly balance detection evasion against functional preservation, requiring domain-specific expertise in both network protocols and attack methodologies to develop effective solutions [6].

5.5. Bridging strategies: From feature space to packet space

The three mapping scenarios identified in Section 5.3 – one-to-many ambiguity, many-to-none infeasibility, and side-effect cascades – collectively define the inverse feature-mapping problem as a fundamental barrier to practical adversarial attacks against NIDS. However, the literature reveals several emerging methodological directions that collectively constitute a research roadmap for bridging this gap. These strategies do not eliminate the inverse mapping challenge, but they reframe it in ways that make practical packet-level attacks increasingly tractable.

5.5.1. Problem-space attack design

The most direct bridge is to abandon feature-space optimisation entirely and formulate adversarial attack generation directly in the packet or problem space. Rather than perturbing a feature vector and then attempting the intractable inverse mapping, problem-space methods operate on raw packet bytes and impose protocol validity as hard constraints during the attack generation process. Deep PackGen exemplifies this approach by formulating packet perturbation as a sequential decision-making problem in which the state space represents raw packet bytes and the action space encompasses protocol-valid perturbation operations [6,7]. The Bellman-equation reward function $Q(s, a) = \mathbb{E}\left[R(s, a) + \gamma \max_{a'} Q(s', a')\right]$ explicitly penalises protocol violations, embedding constraint satisfaction within the optimisation objective rather than treating it as a post-hoc filter. This design principle achieves 66.4% average evasion success while generating functionally valid packets — a substantially more meaningful metric than the unconstrained feature-space success rates exceeding 95% that characterise most published attacks. The key research implication is that future attack evaluation should mandate packet-level validation as a primary metric, with feature-space success treated as an upper bound rather than a deployment estimate.

5.5.2. Differentiable constraint layers

For attacks that must retain gradient-based optimisation – due to architectural or performance requirements – a second bridging strategy encodes network protocol constraints as differentiable penalty terms within the attack objective. Formally, the standard adversarial optimisation $\min_\delta \mathcal{L}(f(x+\delta), y)$ subject to $\|\delta\|_p \leq \epsilon$ is extended to $\min_\delta \mathcal{L}(f(x+\delta), y) + \sum_i \lambda_i \, g_i(x+\delta)$, where each constraint function $g_i(\cdot)$ captures a specific protocol requirement – checksum consistency, flag combination validity, port range bounds, or TCP sequence number relationships – and λ_i controls the penalty weight [11,12]. This approach preserves gradient flow through the attack generation process while progressively guiding perturbations toward the feasible region defined by network protocol semantics. The TransfficFormer framework demonstrates that transformer-based architectures with constraint-aware loss functions can generate evasive traffic that maintains header integrity, reducing the many-to-none infeasibility rate while retaining meaningful evasion capability [11]. Critically, this strategy converts the hard constraint satisfaction problem into a soft optimisation problem, making it amenable to standard deep learning toolchains without requiring specialised constraint solvers.

5.5.3. Generative inverse mapping

The third bridging strategy circumvents the inverse mapping problem by learning a generative model that maps from a latent attack-intent space directly to valid network packets, bypassing feature-space perturbation altogether. Generative Adversarial Networks trained on real network traffic – including IDSGAN [51], CTGAN [75], and Deep PackGen [7] – implicitly learn the joint distribution of valid packet-level features, ensuring that generated samples reside within the feasible region of network protocol space by construction. Rather than solving $F(p+\Delta p) - F(p) = \Delta f$, these methods learn a mapping $G : z \to p^*$ where p^* satisfies both evasion objectives and protocol constraints, with the discriminator enforcing distributional realism as a proxy for packet validity. This approach is particularly effective against the side-effect mapping problem, as the generative model captures inter-feature correlations (flow duration, packets-per-second, inter-arrival time means) as statistical regularities in the training distribution rather than as explicit constraint equations. The fundamental limitation is that generative methods sacrifice fine-grained control over specific feature perturbations in exchange for distributional validity, making them better suited for evasion tasks than for targeted feature manipulation.

Together, these three strategies define a complementary research agenda: problem-space design provides the conceptually cleanest solution but requires architectural commitment; differentiable constraints offer a practical near-term bridge for existing gradient-based frameworks; and generative inverse mapping provides the strongest distributional validity guarantees at the cost of directional control. The common thread is that all three approaches treat protocol compliance as a first-class design constraint rather than an afterthought, which is the essential shift the field must make to close the gap between laboratory success rates and operational attack viability.

6. Defense mechanisms and countermeasures

The defense landscape has evolved through sophisticated countermeasures including adversarial training, ensemble methods, and moving target defense strategies. Table 4 provides a comprehensive performance analysis of defense mechanisms under adversarial conditions, comparing clean accuracy, adversarial accuracy, computational overhead, training time increase, inference latency, and real-time viability across multiple defense categories including adversarial training, detection-based defenses, feature engineering, moving target defense, federated learning, ensemble methods, certified defenses, game-theoretic approaches, input preprocessing, and lightweight defenses. Fig. 6 illustrates four defense architecture frameworks directly corresponding to Section 6 subsections and the Table 4 defense categories. The Adversarial Training Architecture maps to Section 6.1 (Table 4: adversarial training rows); the Ensemble Defense Architecture to Section 6.6 (Table 4: ensemble methods); the MTD Architecture to Section 6.4 (Table 4: moving target defense rows); and the Hybrid Adaptive Defense Architecture to Section 6.10, which synthesises all three into a unified framework that dynamically selects and weights component defenses based on observed threat patterns. Table 4 quantifies the performance trade-offs of each architecture under adversarial conditions.

6.1. Adversarial training approaches

Adversarial training represents the most extensively studied defense mechanism against adversarial attacks, fundamentally altering the training paradigm to expose models to adversarial examples during the learning process [32]. This approach operates on the principle that models trained with adversarial examples develop robustness against similar perturbations encountered during deployment, effectively hardening decision boundaries against adversarial manipulation [8].

6.1.1. Enhanced adversarial training methodologies

The ADV_NN curriculum-based approach exemplifies sophisticated adversarial training implementations, maintaining over 80% accuracy under PGD and FGSM attacks at perturbation levels of $\epsilon = 0.15$ while preserving 86.04% clean data accuracy [37]. This framework integrates gradient alignment mechanisms for feature smoothing, curriculum learning strategies for progressive model hardening, and multi-attack exposure protocols that encompass diverse adversarial generation techniques. The mathematical formulation combines adversarial loss with regularization terms to balance robustness against accuracy degradation: $L_{total} = L_{clean} + \lambda_{adv} L_{adversarial} + \lambda_{reg} L_{regularization}$, where hyperparameter tuning determines optimal trade-offs between competing objectives [13].

6.1.2. Multi-phase ensemble defense frameworks

The most promising adversarial training advancement involves multi-phase ensemble architectures that combine adversarial training with complementary defense mechanisms [13]. These frameworks integrate adversarial training with label smoothing techniques that reduce model overconfidence, Gaussian augmentation for input noise resilience, and Bayesian-optimized weighted voting schemes for ensemble prediction aggregation. The mathematical optimization objective becomes: $\theta^* = \arg\min[\lambda_1 L_{adv}(\theta) + \lambda_2 L_{gauss}(\theta) + \lambda_3 L_{smooth}(\theta)]$, achieving 98.78% accuracy under controlled adversarial conditions [13]. However, these sophisticated approaches face significant deployment challenges, requiring 50–60% additional training time and exhibiting brittleness when individual ensemble components are compromised, with performance degradation exceeding 15% when any single defense mechanism fails [38].

6.2. Detection-based defenses

Detection-based defenses identify adversarial examples prior to classification by exploiting distinguishable characteristics they introduce into traffic representations, operating independently of the primary NIDS classifier [76,77].

6.2.1. Statistical anomaly detection

Statistical anomaly detection identifies adversarial inputs through distributional divergence from benign traffic baselines. MagNet exemplifies this paradigm, constructing a benign traffic manifold and flagging inputs whose statistical signatures exceed calibrated deviation thresholds [78]. Operating on distributional distance metrics rather than per-sample reconstruction, MagNet is computationally lightweight and classifier-agnostic. Its primary limitation is distributional sensitivity: adversaries employing statistical-matching perturbations that preserve feature distributions can achieve bypass rates exceeding 70% [79].

6.2.2. Reconstruction-based detection

Reconstruction-based detection trains autoencoders exclusively on benign traffic, flagging inputs whose reconstruction error $RE(x) = \|x - \text{Dec}(\text{Enc}(x))\|^2$ exceeds the benign threshold [80]. NIDS-DA focuses on non-functional attributes that adversarial perturbations alter while preserving attack functionality, achieving substantial F1-score improvement without adversarial training [81]. RAIDS applies the same reconstruction-error principle at the packet level, capturing perturbation signatures invisible to threshold-based classifiers [82]. Both systems are vulnerable to manifold-aware adversaries who constrain perturbations to remain within the reconstructable space [69].

6.3. Robust feature engineering

Robust feature engineering defends by prioritising network characteristics that are inherently resistant to adversarial manipulation due to protocol constraints or structural relationships [11,12].

6.3.1. Protocol-aware feature selection

Protocol-aware feature selection identifies manipulation-resistant features through systematic analysis of network protocol semantics. The Perturbability Score (PS) framework quantifies feature vulnerability as $PS(f_i) = \alpha \cdot \text{manipulability}(f_i) + \beta \cdot \text{constraint_violation}(f_i)$, achieving a 35% detection accuracy improvement over standard feature selection [43]. Features bound to protocol semantics – TCP flags, port numbers, connection state – resist manipulation because modification violates protocol specifications. Limitation: correlation-aware attackers preserving TCP/UDP compliance while targeting statistical aggregations achieve bypass rates exceeding 87% [38].

6.3.2. Graph-based structural features

Graph-based features capture multi-flow network topology relationships that transcend individual flow statistics, providing inherent adversarial resistance because simultaneous manipulation of coordinated flows is substantially harder than single-flow modification [83]. E-GraphSAGE with Scattering Transform maintains 50% better F1-scores under adversarial attacks than statistical flow-feature classifiers [84]. The message-passing update captures DDoS coordination and port-scanning sequences. Limitation: graph construction scales as $O(|V| + |E|)$ per iteration, creating bottlenecks in high-throughput environments [85].

6.3.3. Protocol constraint-aware defense

Protocol constraint-aware defenses enforce network protocol validity rules as detection signals, flagging traffic that violates field dependencies such as TCP ACK/SEQ relationships, IP length consistency, or discrete-field enumeration constraints [9,11]. Because adversarial perturbations that modify protocol-constrained fields produce invalid packets detectable at the infrastructure layer, protocol constraint monitoring provides a defense layer orthogonal to learned classifiers. Limitation: adversaries employing constraint-satisfying perturbation methods bypass this defense entirely, shifting the burden to statistical and correlation-based detection [12].

6.3.4. Robust feature aggregation

Robust aggregation combines feature representations across multiple extraction pipelines – statistical, protocol, and temporal – such that corrupting one representation is insufficient to fool the aggregated decision [12]. Ensemble feature aggregation imposes higher manipulation cost on adversaries: to evade detection, perturbations must simultaneously corrupt all feature views while maintaining packet validity across all protocol layers. Limitation: aggregation increases feature dimensionality and computational overhead, and adversaries with access to the full aggregation structure can target the weakest component pipeline [38].

6.3.5. Graph-based structural features

Graph-based structural feature engineering represents a paradigmatic advancement in adversarial robustness by capturing network topology relationships that transcend individual flow characteristics [83].

The E-GraphSAGE framework with Scattering Transform integration demonstrates remarkable adversarial resilience, maintaining 50% better F1-scores under adversarial attacks compared to traditional machine learning approaches that rely solely on statistical flow features [84]. Graph Neural Networks employ message passing mechanisms governed by $h_v^{(l+1)} = UPDATE(h_v^l, AGGREGATE(\{h_u^l : u \in N(v)\}))$, where structural invariance captures multi-flow attack patterns including DDoS coordination, port scanning sequences, and brute force distribution across network topology [109]. This approach provides inherent resistance to perturbation-based attacks as adversaries would need to manipulate multiple coordinated flows simultaneously while maintaining attack coordination–a substantially more complex undertaking than single-flow modifications. However, scalability challenges emerge in large network deployments where graph construction and message passing computation scales as $O(|V| + |E|)$ per iteration, potentially creating processing bottlenecks in high-throughput environments [85].

6.4. Moving target defense (MTD)

Moving Target Defense (MTD) invalidates adversarial reconnaissance by continuously modifying the attack surface [110,111]. The three MTD strategies differ in trigger mechanism and change type.

6.4.1. Dynamic configuration shuffling

Dynamic configuration shuffling periodically reconfigures network parameters – IP address assignment, port rotation, service location – through SDN-managed entropy maximisation $H(X) = -\sum p(x_i) \log p(x_i)$, ensuring adversarial reconnaissance maps become stale before exploitation [112]. Unlike confidence-based and topology MTD, dynamic shuffling is proactive and schedule-driven rather than threat-reactive, providing continuous disruption at low inference overhead. Limitation: effective deployment requires SDN-capable hardware currently absent from 95% of surveyed organisations [113].

6.4.2. Confidence-based MTD

Confidence-based MTD triggers reconfiguration reactively: when classifier confidence $p(y|x, D)$ drops below a predetermined threshold, the system activates configuration changes to restore detection reliability [114,115]. Unlike dynamic shuffling, this approach responds to evidence of active probing, reducing unnecessary reconfiguration overhead. Key differentiator from Topology Mutation: it changes classifier parameters, not network structure. Limitation: centralised MTD coordination becomes a high-value target and single point of failure in distributed deployments [116].

6.4.3. Topology mutation

Topology mutation alters the structural network graph itself – routing paths, node roles, link topology – rather than reconfiguring parameters within a fixed structure [110]. This provides qualitatively stronger disruption than dynamic shuffling or confidence-based MTD: adversaries cannot exploit persistent attack paths even after adapting to parameter changes, because the structural attack surface has changed. Limitation: topology mutation requires overlay network control or SDN fabric support and incurs higher reconfiguration latency than parameter-level MTD, restricting deployment to security-prioritised network segments.

6.5. Federated learning defenses

Federated learning defenses enable collaborative adversarial robustness across distributed deployments without centralising traffic data, defending against poisoning attacks on the distributed training process [117,118]. The four strategies differ in aggregation mechanism.

6.5.1. SecFedNIDS

SecFedNIDS defends against poisoning attacks through dual-layer gradient analysis: server-side gradient-based parameter selection identifies poisoned client updates, while layer-wise relevance propagation extracts clean data contributions [94]. This gradient-distribution analysis achieves substantial accuracy improvement under poisoning attacks on UNSW-NB15 and CICIDS2018. Limitation: server-side-only operation allows client-side evasion strategies that maintain normal validation signatures [73].

6.5.2. Personalised federated learning (pFL-IDS)

pFL-IDS addresses heterogeneous NIDS deployments by clustering clients via cosine similarity, isolating malicious participants and adapting the global model to local network characteristics [119,120]. Unlike SecFedNIDS (gradient-analysis, server-side), pFL-IDS is client-clustering-based and provides cross-attack robustness across diverse organisational environments. Limitation: the personalisation process may leak organisational network pattern information through client similarity scores [73].

6.5.3. Byzantine robust federated learning

Byzantine Robust FL applies Krum aggregation – selecting the gradient update geometrically closest to its neighbours and discarding outliers corresponding to poisoned clients – providing provable Byzantine fault tolerance without server-side training [96]. Unlike SecFedNIDS (gradient distribution analysis) and pFL-IDS (clustering), Krum operates purely on geometric distance in update space. Limitation: Krum assumes a bounded fraction of Byzantine clients; coordinated attacks exceeding this fraction can still corrupt aggregation.

Table 4
Defense mechanism performance under adversarial conditions.

Defense Category	Method	Mathematical Foundation	Dataset	Clean Accuracy (%)	Adversarial Accuracy (%)	Computational Overhead	Training Time Increase	Inference Latency	Real-time Viability	Attack Types Defended	Key Limitations	Citations
Adversarial Training	**ADV_NN**	$L_{total} = L_{clean} + \lambda_{adv} L_{adversarial} + \lambda_{reg} L_{regularization}$	CSE-CIC-IDS2018	86.04	80+ (ϵ = 0.15)	50–60%	50–60%	15.97s	High	FGSM, PGD	Brittleness to adaptive attacks	[37]
	Multi-Phase Ensemble	$\theta^* = \arg\min[\lambda_1 L_{adv}(\theta) + \lambda_2 L_{gauss}(\theta) + \lambda_3 L_{smooth}(\theta)]$	CIC-IDS2017	98.78	98.78 (weighted)	3–10x	200–300%	50–100s	Medium	Multiple types	Component failure vulnerability	[13]
	Curriculum Learning	Progressive hardening with gradient alignment	NSL-KDD	92.5	75–85	80–100%	100–150%	25–40s	Medium	PGD, C&W	Curriculum design complexity	[37]
	TRADES	$\min_\theta E[L(f_\theta(x), y) + \beta \max_{\|\delta\| \le \epsilon} L(f_\theta(x), f_\theta(x+\delta))]$	UNSW-NB15	88.9	72–82	75–90%	120–180%	20–35s	Medium	Gradient-based	Trade-off parameter sensitivity	[86]
	MART	$\min_\theta E[\max(\alpha(L_{KL}(f_\theta(x+\delta)) \| f_\theta(x)) - L_{CE}), L_{CE})]$	CIC-IDS2017	90.3	78–88	85–110%	150–200%	30–45s	Medium	Multiple gradients	Overfitting to specific attacks	[87]
	Ensemble Adversarial Training (EAT)	Multi-model adversarial example generation	Multi-dataset	94.2	85–93	200–300%	250–400%	60–120s	Low	Transfer attacks	High computational cost	[66]
Detection -Based	**NIDS-DA**	Deep autoencoder analysis reconstruction error analysis	CIC-IDS2017	95.1	85–92	120–150%	80–120%	150–200 ms	Low	Statistical anomalies	Reconstruction time overhead	[81]
	RAIDS	$\text{Score}(x) = -\|\text{Encoder}(\text{Decoder}(x)) - x\|_2$	InSDN	93.8	80–88	100–130%	70–100%	120–180 ms	Low	Reconstruction-based	Adaptive attack vulnerability	[82]
	Statistical Anomaly	$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$	Various	89.2	70–85	40–60%	20–40%	50–80ms	High	Distribution shifts	Statistical matching attacks	[88]
	MagNet	Two-pronged defense with detector and reformer	Multi-dataset	87.6	72–85	90–120%	60–90%	100–150 ms	Medium	Various gradients	Detector bypass techniques	[89]
Feature Engineering	**Perturbability Score**	$PS(f_i) = \alpha \cdot \text{manipulability}(f_i) + \beta \cdot \text{constraint_violation}(f_i)$	Multi-dataset	94.7	88–94	15–25%	10–20%	5–15s	Very High	Protocol-aware	Limited feature coverage	[90]
	Graph-based Features	$h_v^{(l+1)} = \text{UPDATE}(h_v^l, \text{AGGREGATE}(\{h_u^l : u \in N(v)\}))$	Various	92.3	85–91	40–70%	50–80%	25–50s	High	Structural attacks	Scalability challenges	[91]
	Protocol Constraints	Domain-specific constraint enforcement	Network flows	89.8	82–89	20–35%	15–30%	10–25s	Very High	Constraint violations	Protocol-specific implementation	[9]
	Robust Aggregation	$\text{TrimmedMean}_\alpha(x_1, \ldots, x_n) = \frac{1}{n-2k} \sum_{i=k+1}^{n-k} x_{(i)}$	Ensemble data	90.5	80–87	30–45%	25–40%	15–30s	High	Outlier manipulation	Trimming parameter tuning	[23]

Table 4
Cont.

Defense Category	Method	Mathematical Foundation	Dataset	Clean Accuracy (%)	Adversarial Accuracy (%)	Computational Overhead	Training Time Increase	Inference Latency	Real-time Viability	Attack Types Defended	Key Limitations	Citations
Moving Target Defense	**Dynamic IP Shuffling**	$H(X) = -\sum_i p(x_i) \log p(x_i)$ (entropy maximization)	SDN testbed	91.2	85–92	200–400%	N/A	100–300ms	Low	Reconnaissance	Infrastructure requirements	[67]
	Confidence-based MTD	$p(y\|x, D) = \int p(y\|x, \theta)p(\theta\|D)d\theta$	Multi-environment	93.6	87–94	150–250%	80–120%	80–150ms	Medium	Adaptive attacks	Coordination overhead	[92]
	Topology Mutation	Systematic network reconfiguration	Enterprise networks	88.9	82–88	300–500%	N/A	200–500ms	Very Low	Persistent threats	High deployment cost	[93]
Federated Learning	**SecFed-NIDS**	$w_{t+1} = w_t - \eta\nabla L(w_t; D_{\text{clean}})$	Multi-organization	94.2	86–93	80–120%	150–250%	60–120ms	Medium	Poisoning attacks	Scalability limitations	[94]
	pFL-IDS	Personalized federated learning with similarity alignment	Distributed NIDS	96.5	88–95	100–150%	200–300%	80–150ms	Medium	Backdoor attacks	Client clustering complexity	[95]
	Byzantine Robust FL	$\text{Krum}(g_1, \ldots, g_n) = \arg\min_{g_i} \sum_{j \in S_i} \|g_i - g_j\|^2$	Federated setup	91.8	83–90	120–180%	180–280%	100–200ms	Low	Byzantine attacks	Communication overhead	[96]
	FedDef	Input perturbation optimization with gradient deviation constraint	KDD99, CIC-IDS2017	98.7, 97.1	95–98	90–140%	120–200%	70–130ms	Medium	Privacy attacks	Parameter sensitivity	[97]
Ensemble Methods	**Voting Ensemble**	$\hat{y} = \arg\max_c \sum_{i=1}^{M} \mathbf{1}_{h_i(x)=c}$	CIC-IDS2017	97.2	91–96	300–400%	100–150%	200–400s	Medium	Transfer attacks	Computational overhead	[98]
	Weighted Ensemble	$\hat{y} = \arg\max_c \sum_{i=1}^{M} w_i \cdot P_i(y = c\|x)$	UNSW-NB15	95.8	89–94	250–350%	120–180%	150–300s	Medium	Diverse attacks	Weight optimization complexity	[99]
	Hierarchical Ensemble	Layered decision with confidence thresholds	Multi-dataset	96.4	90–95	400–600%	200–300%	300–600s	Low	Complex attacks	Deep hierarchy limitations	[100]
	TIKI-TAKA	Multi-phase ensemble with adaptive weights	Advanced datasets	99.1	98–99	500–800%	300–500%	400–800s	Very Low	Cutting-edge attacks	Extreme resource requirements	[101]
Certified Defenses	**Randomized Smoothing**	$g(x) = \arg\max_c P(f(x + \epsilon) = c), \epsilon \sim N(0, \sigma^2 I)$	Theoretical	85.4	75–85 (certified)	1000–5000%	500–1000%	1–10s	Very Low	ℓ_2 perturbations	Gaussian noise requirements	[102]
	Interval Bound Propagation	$\text{IBP}(x, \epsilon) = [L^{(k)}, U^{(k)}]$ layer propagation	Limited datasets	82.1	70–82 (certified)	2000–8000%	800–1500%	5–30s	Very Low	ℓ_∞ perturbations	Conservative bounds	[103]
	MARS (Multi-order Adaptive)	Multi-order smoothing with adaptive parameters	Network traffic	87.9	78–88 (certified)	800–1500%	400–800%	2–15s	Very Low	Heterogeneous features	Feature type limitations	[81]

Table 4
Cont.

Defense Category	Method	Mathematical Foundation	Dataset	Clean Accuracy (%)	Adversarial Accuracy (%)	Computational Overhead	Training Time Increase	Inference Latency	Real-time Viability	Attack Types Defended	Key Limitations	Citations
Game-Theoretic	**Stackelberg Games**	Nash equilibrium with defender advantage	Simulated	89.7	78–86	500–1000%	300–600%	500ms–2s	Very Low	Strategic attacks	Computational complexity	[104]
	Zero-Sum Games	$\min_\theta \max_\delta E[L(f_\theta(x+\delta), y)]$	Theoretical	87.3	75–85	800–1500%	400–800%	1–5s	Very Low	Adaptive adversaries	PPAD-complete solutions	[105]
Lightweight Defenses	**Quantized Defense**	8-bit/16-bit model quantization	Edge devices	96.8	88–95	50–80%	30–60%	10–25ms	Very High	Precision-based attacks	Accuracy trade-offs	[106]
	FPGA Implementation	Hardware pipeline processing	NetFPGA	98.22	91–97	Hardware-fixed	N/A	5–15ms	Extremely High	Various attacks	Hardware limitations	[107]
	Bio-Inspired Optimization	GMGWO + LDA feature selection	NSL-KDD	96.08	89–95	40–70%	50–90%	15–40ms	High	Feature manipulation	Parameter sensitivity	[108]

6.5.4. FedDef

FedDef simultaneously defends against Byzantine gradient attacks and inference attacks by combining differential privacy noise injection with gradient anomaly filtering [97]. This dual-mechanism design is unique among Table 4 FL methods in addressing both attack surfaces concurrently. Limitation: differential privacy noise reduces model utility, requiring careful ϵ-calibration to balance privacy guarantees against detection accuracy.

6.6. Ensemble defense methods

Ensemble defense methods achieve adversarial robustness through classifier diversity, aggregating predictions from multiple independently trained classifiers so that a single perturbation cannot simultaneously fool all members. Voting ensembles select the majority class across M classifiers; weighted ensembles assign confidence-calibrated weights that down-grade classifiers exhibiting anomalous outputs; hierarchical ensembles apply layered confidence thresholds to further filter uncertain predictions [13,99,100]. TIKI-TAKA advances this with multi-phase adaptive weighting that responds to observed attack patterns, achieving the highest adversarial accuracy among all Table 4 defense categories [101]. The defining trade-off is component vulnerability: performance degrades by over 15% when any single member is compromised, and training overhead is prohibitive for real-time deployment [13].

6.7. Certified defense approaches

Certified defenses provide mathematically provable robustness bounds rather than empirically observed robustness. *Randomised Smoothing* constructs a certified classifier by selecting the class most probable under Gaussian-perturbed inputs, issuing guarantees only when the margin exceeds a statistical confidence threshold [102]. *Interval Bound Propagation (IBP)* computes exact worst-case output bounds layer-by-layer, ensuring no perturbation within the certified region changes the predicted class [103]. *MARS* extends Randomised Smoothing with adaptive noise scheduling that adjusts σ to local feature geometry, improving certified accuracy on heterogeneous network traffic features. The fundamental limitation is the accuracy–certification trade-off: certified methods record the lowest clean accuracy in Table 4, and inference latency renders them suitable only for offline forensic analysis [81].

6.8. Game-theoretic defense frameworks

Game-theoretic frameworks model attacker and defender as strategic agents, applying equilibrium analysis to derive defense policies robust against rational adversaries. Stackelberg formulations treat the interaction as a sequential leader-follower game: the defender commits first and the adversary responds optimally, yielding the minimax objective equivalent to adversarial training but grounded in rational-agent theory [104]. Zero-sum formulations treat adversarial gain as direct defensive loss, enabling linear programming to derive provably optimal mixed-strategy policies. The primary limitation is PPAD-completeness: computing Nash equilibria for the general adversarial game is computationally intractable, requiring strategic-space simplifications that may not hold against adaptive real-world adversaries [105].

6.9. Lightweight defense mechanisms

Lightweight defenses prioritise deployment feasibility in resource-constrained NIDS environments where adversarial training overhead is impractical [106]. Quantised Defense replaces full-precision weights with 8-bit or 16-bit representations, providing implicit regularisation that reduces adversarial sensitivity while cutting inference latency by 60–80% [121]. FPGA Implementation achieves deterministic hardware-pipeline processing (5–15ms latency), making it the only Table 4 category rated Extremely High for real-time viability. Bio-Inspired Optimisation (GMGWO + LDA) applies evolutionary feature selection to identify adversarially robust feature subsets without gradient computation. The unifying limitation is adaptability: FPGA logic requires re-synthesis to update defense rules, and bio-inspired methods require per-deployment hyperparameter calibration [107].

6.10. Hybrid adaptive defense architecture

The Hybrid Adaptive Defense Architecture integrates adversarial training (Section 6.1), ensemble aggregation (Section 6.6), and MTD reconfiguration (Section 6.4) within a meta-controller that continuously weights component defenses according to their real-time evasion-detection accuracy: $\theta^* = \arg\min_\theta \sum_k \lambda_k \mathcal{L}_k(\theta)$, where λ_k are updated by a reinforcement learning policy observing detection outcomes [101]. Adversarial training provides a baseline robustness floor; ensemble aggregation ensures that adversarial examples must defeat all members simultaneously; MTD reconfiguration denies adversaries a stable attack surface [92,93]. The architecture achieves the highest adversarial accuracy of any deployed defense category but imposes combined overhead that restricts real-time viability to dedicated security hardware [91].

7. Adaptive adversarial strategies

7.1. The evolution to adaptive attacks

Adaptive adversarial attacks represent a fundamental paradigm shift from static threat models to dynamic strategies that continuously evolve based on defense mechanisms and system responses [122]. This evolution emerged as researchers recognized that traditional static adversarial examples, while effective in laboratory settings, fail to capture the sophisticated capabilities of real-world adversaries who possess the ability to observe defensive responses and modify their attack strategies accordingly [123]. The critical distinction lies in the feedback mechanism: while static attacks operate with fixed perturbation strategies, adaptive attacks leverage real-time intelligence gathering to circumvent defensive measures through iterative refinement processes [48].

Research demonstrates that adaptive attacks achieve devastating effectiveness against static defense mechanisms, with documented cases showing 50%+ degradation in NIDS accuracy while maintaining near-perfect stealth capabilities [37]. The mathematical foundation involves dynamic optimization where attack parameters evolve according to: $\theta_{t+1} = \theta_t - \alpha \nabla_\theta L_{defense}(\theta_t)$, enabling adversaries to systematically probe defensive boundaries and identify optimal evasion strategies. This adaptive capability fundamentally challenges the security assumptions underlying current ML-based NIDS deployments, as defensive systems designed for static threat models become systematically vulnerable to adversaries who can observe, learn, and adapt their strategies in real-time [4]. The implications extend beyond technical considerations to encompass fundamental game-theoretic dynamics where defenders must anticipate not only current attack capabilities but also the adaptive potential of sophisticated adversaries [124].

7.2. Dynamic feedback mechanisms

Dynamic feedback mechanisms represent the core innovation enabling adaptive adversarial attacks to systematically probe and circumvent defense systems through iterative intelligence gathering and strategy refinement [125]. These mechanisms exploit the fundamental asymmetry in cybersecurity environments where defensive systems must respond to inputs while inadvertently revealing information about their internal decision-making processes through observable outputs, timing characteristics, and system resource consumption patterns [34].

7.2.1. Real-time defense learning

Real-time defense learning exploits side-channel indicators including response time analysis for decision boundary mapping, CPU usage patterns for model complexity inference, and memory consumption variations for architecture fingerprinting [71]. Sophisticated implementations employ statistical correlation analysis to establish relationships between input modifications and system responses, enabling adversaries to build surrogate models that approximate target NIDS behavior without requiring direct model access. The mathematical framework utilizes finite difference approximation for gradient estimation: $\hat{\nabla}_x f(x) \approx \frac{f(x+\delta e_i) - f(x-\delta e_i)}{2\delta}$ for each coordinate direction e_i, where query efficiency becomes critical for practical deployment [46]. Advanced adversaries implement adaptive query strategies that minimize detection risk while maximizing information extraction, achieving effective gradient approximation with logarithmic query complexity relative to feature dimensionality. However, defensive countermeasures including query rate limiting, response randomization, and differential privacy mechanisms can significantly degrade the effectiveness of real-time learning approaches, forcing adversaries to balance information gathering speed against detection avoidance [66].

7.2.2. Change-point detection

Change-point detection algorithms enable adversaries to identify moments when NIDS behavior changes significantly in response to traffic modifications, revealing defensive adaptation strategies and vulnerability patterns [126]. Binary Segmentation algorithms with different models utilize radial basis functions for non-linear variation detection and L_2 norm analysis for linear shift identification, mathematically formulated as: $\hat{\tau} = \arg\max_{1 \le t \le n} |S_t|$, where S_t represents the cumulative sum statistic. The approach combines segment-based regression analysis using Ordinary Least Squares with Variance Inflation Factor filtering to preserve feature interdependencies, followed by statistical significance testing to identify truly causal relationships between input modifications and defensive responses. Three-step causal analysis enables adversaries to distinguish correlation from causation, ensuring attack strategies target genuinely vulnerable system components rather than spurious correlations. However, implementation complexity increases substantially in high-dimensional feature spaces where multiple simultaneous changes may obscure causal relationships, while sophisticated defensive systems can employ adversarial training specifically designed to minimize observable behavioral changes under attack [86].

7.3. Game-theoretic frameworks

Game-theoretic frameworks provide mathematical rigor to the strategic interactions between adaptive adversaries and intelligent defensive systems, modeling cybersecurity as a dynamic competition where both parties continuously optimize their strategies based on opponent behaviors [127]. These frameworks acknowledge that network security operates within an inherently adversarial environment where rational actors pursue conflicting objectives, necessitating formal mathematical models that capture the strategic decision-making processes underlying both attack and defense strategies [128].

7.3.1. Stackelberg game models

Stackelberg game models provide mathematical frameworks for proactive security strategies where defenders move first and attackers respond optimally to observed defensive configurations [129]. This leader-follower paradigm enables defenders to anticipate adversarial responses and implement strategies that account for optimal attacker adaptation, mathematically formulated as: $\max_{s_d} \min_{s_a} U_d(s_d, s_a)$, where s_d represents defensive strategies and s_a denotes adversarial responses. The framework assumes perfect information for the follower (attacker) who observes defensive commitments before choosing optimal responses, enabling sophisticated threat modeling that captures realistic attack scenarios where adversaries conduct reconnaissance before launching attacks. Advanced implementations incorporate uncertainty through Bayesian Stackelberg games that model incomplete information about attacker types and capabilities: $EU_d(s_d) = \sum_\theta p(\theta) U_d(s_d, BR_a(s_d, \theta))$, where θ represents attacker types and BR_a denotes best response functions. However, computational complexity grows exponentially with strategy space dimensionality, while the assumption of rational adversaries with complete information rarely holds in practice, limiting applicability to simplified network security scenarios [130].

7.3.2. Nash equilibrium strategies

Nash equilibrium strategies represent stable solution concepts where neither attackers nor defenders can unilaterally improve their payoffs by changing strategies, mathematically defined by the condition: $U_d(s_d^*, s_a^*) \ge U_d(s_d, s_a^*)$ and $U_a(s_d^*, s_a^*) \ge U_a(s_d^*, s_a)$ for all alternative strategies [131]. Multi-agent Markov game frameworks enable continuous strategy adaptation through reinforcement learning algorithms that converge toward equilibrium solutions, combining Bayesian Strong Stackelberg approaches with WoLF (Win-or-Learn-Fast) algorithms to obtain equilibrium points in dynamic environments [132]. The mathematical formulation involves policy gradient methods: $\nabla_\theta J(\theta) = \mathbb{E}[\nabla_\theta \log \pi_\theta(a|s) Q^\pi(s, a)]$, enabling adaptive strategy evolution. However, Nash equilibrium computation is PPAD-complete, making real-time application computationally intractable for realistic network security scenarios. Furthermore, multiple equilibria often exist without clear selection criteria, while the assumption of simultaneous strategy selection contradicts the sequential nature of most cybersecurity interactions [105].

7.4. Meta-learning approaches

Meta-learning approaches address the fundamental challenge of rapid adaptation to novel adversarial threats by developing algorithms capable of learning how to learn new attack patterns efficiently with minimal training data [133]. This paradigm becomes increasingly critical in network security contexts where zero-day attacks and novel threat vectors emerge continuously, necessitating defensive systems that can quickly adapt to previously unseen adversarial strategies without requiring extensive retraining on large datasets [134].

7.4.1. Model-agnostic meta-learning (MAML)

Model-Agnostic Meta-Learning frameworks enable rapid adaptation to new IoT environments and attack scenarios through gradient-based optimization that learns optimal initialization parameters for fast adaptation [134]. The mathematical foundation involves bi-level optimization where the meta-objective optimizes for parameters that enable rapid task-specific adaptation: $\theta^* = \arg\min_\theta \sum_{T_i \sim p(T)} L_{T_i}(\theta - \alpha \nabla_\theta L_{T_i}(\theta))$, where T_i represents individual attack scenarios and α controls adaptation step size. Empirical evaluations demonstrate that MAML-based NIDS achieve 85% detection rates for zero-day attacks while maintaining cross-domain applicability across heterogeneous network environments. The framework excels in federated learning scenarios where organizations can collaborate on meta-learning without sharing sensitive data, enabling knowledge transfer across different network topologies and attack landscapes. However, computational requirements scale quadratically with model complexity due to second-order gradient

computations, while convergence properties remain theoretically unguaranteed in non-convex optimization landscapes typical of deep learning applications [135].

7.4.2. Continuous learning systems

Continuous learning systems implement online learning algorithms that provide immediate adaptation to emerging threats while maintaining knowledge of previously encountered attack patterns to prevent catastrophic forgetting [136]. These systems employ elastic weight consolidation techniques: $L(\theta) = L_B(\theta) + \sum_i \frac{\lambda}{2} F_i(\theta_i - \theta_{A,i})^2$, where F_i represents Fisher information importance weights that preserve critical parameters from previous tasks. Advanced implementations utilize experience replay mechanisms and progressive neural architecture expansion to accommodate new threat categories without degrading performance on established attack types. The approach enables real-time adaptation with incremental training that ensures continuous improvement without complete system redesign, achieving adaptation cycles measured in minutes rather than hours required for traditional retraining approaches. However, memory requirements grow linearly with the number of encountered threat types, while determining optimal replay buffer sizes and importance weight calculations remains challenging in dynamic network environments where threat distributions shift continuously [137].

8. Dataset analysis and evaluation challenges

8.1. Current dataset landscape

The evaluation of adversarial attacks against NIDS relies heavily on benchmark datasets that exhibit significant limitations in capturing the complexity and diversity of real-world network environments [149]. Table 5 provides a comprehensive analysis of dataset characteristics and cross-validation performance, comparing primary benchmark datasets (CIC-IDS2017, UNSW-NB15, CSE-CIC-IDS2018, NSL-KDD), emerging datasets, and specialized datasets across multiple dimensions including records, features, attack types, benign traffic percentage, within-dataset accuracy, cross-dataset accuracy, adversarial robustness, temporal stability, data quality issues, infrastructure realism, and primary attack types.These datasets fundamentally shape research directions and conclusions, yet systematic analysis reveals critical gaps between laboratory evaluation conditions and operational deployment scenarios that compromise the validity of adversarial robustness assessments [150].

8.1.1. Primary benchmark datasets

Five core datasets dominate adversarial NIDS research, establishing de facto evaluation standards despite inherent limitations [21]. CIC-IDS2017 has emerged as the most widely adopted benchmark, featuring 2.8 million flows across 83 features with comprehensive attack coverage spanning DDoS, botnet activities, brute force attacks, and web-based intrusions [138]. UNSW-NB15 provides 257,000 flows with 49 features across 9 modern attack categories, valued for its balanced class distribution that facilitates robust statistical analysis [140]. CSE-CIC-IDS2018 extends the evaluation scope with 16 million flows captured over 10 days, incorporating realistic AWS infrastructure deployment scenarios [139]. NSL-KDD, despite acknowledged limitations including outdated attack patterns and unrealistic traffic distributions, continues serving as a baseline comparison platform for algorithmic validation [141]. However, systematic cross-dataset evaluation reveals catastrophic generalization failures, with models achieving 95%+ accuracy within datasets dropping to 30–40% accuracy across datasets–performance "largely commensurate with random chance"–indicating fundamental distributional mismatches rather than genuine learning [21].

8.1.2. Emerging datasets (2022–2025)

Recent dataset development addresses limitations of primary benchmarks by targeting modern network architectures, domain-specific threats, and multimodal feature integration [145]. UM-NIDS (2024) advances cross-dataset validation through multimodal unification – combining flow data, packet payloads, and temporal features across established benchmarks into standardised 8-24GB formats [143]. CICIoMT2024 (2024) provides healthcare-IoT adversarial evaluation through 40 IoMT devices generating 18 attack types, enabling device lifecycle profiling unavailable in generic datasets [144]. NF-UQ-NIDS (2024, 75.99M records) introduces 53 extended NetFlow features including 10 temporal dimensions that support time-series adversarial analysis [145]. CIC-DDoS2019 (2019) offers focused evaluation with 12 DDoS variants across application and transport layers, exposing classifier behaviour under high-volume volumetric attacks [151]. 5G-NIDD (2023) captures 5G-specific traffic to address the fundamental mismatch between legacy benchmarks and next-generation network slicing attack surfaces [146]. TON-IoT (2020, 461K records, 44 features) provides heterogeneous IoT and IIoT telemetry spanning seven attack categories, widely adopted for federated and edge NIDS evaluation [147]. Collectively, these datasets expand adversarial evaluation beyond laboratory Ethernet environments, though cross-dataset transferability remains limited by domain specificity and inconsistent labelling methodologies [22].

8.2. Critical dataset limitations

Contemporary adversarial NIDS research suffers from systematic dataset limitations that compromise the validity and generalisability of research findings [152].

Table 5
Dataset characteristics and cross-validation performance.

Dataset	Year	Records	Features	Attack Types	Benign Traffic (%)	Within-Dataset Accuracy (%)	Cross-Dataset Accuracy (%)	Adversarial Robustness (%)	Temporal Stability	Data Quality Issues	Infrastructure Realism	Primary Attack Types	Citations
Primary Benchmark Datasets													
CIC-IDS2017 (Source link)	2017	2.8M	83	14 categories	75.8	99.2 ± 0.3	42.1 ± 8.7	15.3 ± 4.2	F1: 0.99→0.61 (1 year)	20% calculation errors, 11–15% label errors, 12.5% duplicate features	Laboratory testbed	DDoS, Botnet, Brute Force, Web attacks	[138], [139]
UNSW-NB15 (Source link)	2015	257K	49	9 categories	56.0	95.8 ± 0.4	34.2 ± 6.9	25.8 ± 5.1	Moderate degradation	Synthetic generation artifacts, Label inconsistencies	Hybrid real/synthetic	Analysis, Backdoor, DoS, Exploits, Fuzzers	[140], [21]
CSE-CIC-IDS2018 (Source link)	2018	16M	80	7 scenarios	83.0	98.5 ± 0.2	30.1 ± 7.3	12.7 ± 3.8	Poor (50% minority errors)	18.7% miscalculated features, CICFlowMeter tool errors	AWS cloud infrastructure	Brute Force, Botnet, DoS/DDoS, Web attacks	[139], [11]
NSL-KDD (Source link)	2009	148K	41	4 categories	53.5	99.7 ± 0.1	28.7 ± 9.2	8.2 ± 2.9	Severely outdated	Unrealistic traffic patterns, No modern attack vectors	Simulated network	DoS, R2L, U2R, Probe	[141], [142]
Emerging Datasets (2022–2025)													
UM-NIDS2024 (Source link)	2024	8-24GB	Multi-modal	Combined from 4 datasets	Variable	96.8 ± 0.5	67.3 ± 4.2	45.2 ± 6.7	Under evaluation	Inherits source limitations, Limited documentation	Standardized format	Flow + Payload + Temporal, Cross-dataset validation enabled	[143]
CICIoMT-2024 (Source link)	2024	Variable	Healthcare-specific	18 attack types	78.5	97.2 ± 0.3	52.8 ± 5.9	38.7 ± 5.4	Recent, limited history	Domain-specific bias, Healthcare environment only	40 IoMT devices	DDoS, DoS, Reconnaissance, MQTT, Spoofing	[144]
UQ NetFlow v3 (Source link)	2024	75.99M	53 extended	Combined sources	Variable	94.6 ± 0.4	48.9 ± 6.1	32.1 ± 4.9	Improved temporal features	NetFlow limitations, No payload information	Multi-source integration	10 temporal features	[145]
CIC-DDoS2019 (Source link)	2019	Variable	DDoS-specific	12 DDoS types	50.0	98.9 ± 0.2	38.7 ± 7.8	22.4 ± 4.6	Single attack focus	Limited attack diversity, DDoS-only scope	Specialized testbed	NTP, DNS, LDAP, SYN, UDP floods	[139]
5G-NIDD (Source link)	2023	Variable	5G-specific	Modern 5G attacks	Variable	Under evaluation	Limited evaluation	Unknown	Very recent	Early development stage, Incomplete validation	5G network infrastructure	Network slicing, Edge attacks	[146]
Specialized Datasets													
ToN-IoT (Source link)	2020	461K	44	IoT-specific	70.2	93.4 ± 0.6	41.2 ± 8.1	28.9 ± 5.7	IoT-focused evaluation	72.5% unrealistic topology, Limited scale	Heterogeneous IoT testbed	XSS, Injection, Password, Backdoor, MITM	[147]
Edge-IIoTSet (Source link)	2022	Variable	Industrial IoT	IIoT attacks	65.8	95.7 ± 0.4	44.8 ± 6.7	31.2 ± 4.8	Domain-specific	Industrial environment bias, Resource constraints	Edge computing focus	SCADA, PLC, Industrial protocols	[148]

8.2.1. Data quality issues

Systematic data corruption affects all major NIDS benchmarks. CICFlowMeter tool errors cause feature calculation mistakes in over 20% of traffic flows across CIC-based datasets; labelling inconsistencies reach 11-15% error rates in CIC-IDS2017 and CSE-CIC-IDS2018, with minority attack classes exhibiting 50%+ incorrect labels. CIC-IDS2017 contains 12.5% duplicate features and 18.7% miscalculated features that introduce spurious correlations [139]. NSL-KDD exhibits unrealistic traffic patterns and entirely outdated attack vectors [141].

8.2.2. Temporal relevance problems

Static datasets become obsolete as network traffic patterns evolve. CIC-IDS2017 models degrade from F1 = 0.99 to 0.61 when tested one year after training; UGR'16 performance falls from 0.96 to 0.687 across months within the same year [139]. Adversarial robustness assessments based on outdated traffic patterns are fundamentally unreliable for contemporary threat landscapes.

8.2.3. Cross-dataset generalization failures

Models achieving 95%+ accuracy within datasets consistently drop to 30-40% accuracy when evaluated across datasets, with adversarial robustness degrading further to 12-25% cross-dataset performance compared to 70%+ within-dataset results [21]. These failures suggest research is overfitting to dataset-specific artefacts rather than developing genuinely robust detection mechanisms [22].

8.3. Adversarial evaluation challenges

Adversarial evaluation methodologies in NIDS research exhibit fundamental limitations that compromise the reliability and practical relevance of robustness assessments [12]. These challenges manifest through inconsistent attack success rate measurements, inadequate transferability analysis, and evaluation protocols that fail to capture the complexity of real-world adversarial scenarios where attackers operate under realistic constraints and defenders employ adaptive countermeasures [10].

8.3.1. Attack success rate variations

Attack success rate measurements vary dramatically based on constraint enforcement levels, creating misleading impressions of adversarial vulnerability that do not reflect practical deployment scenarios [11]. Unconstrained attacks operating in pure feature space achieve near-perfect success rates approaching 100%, yet produce approximately 80% invalid packets that would be automatically rejected by network infrastructure [12]. Protocol-aware attacks that maintain basic TCP/UDP compliance achieve only 25% success rates while producing 75% valid packets, demonstrating the substantial impact of realistic constraints on attack effectiveness. Complete constraint enforcement, including semantic preservation and protocol compliance requirements, reduces attack success rates to less than 5% while achieving 95% packet validity [9]. The validity-effectiveness trade-off reveals a fundamental limitation where improvements in packet validity consistently reduce attack effectiveness, suggesting that many published attack success rates substantially overstate practical vulnerability. Hybrid evaluation approaches that balance constraint satisfaction with attack optimization achieve moderate success rates around 40% with 85% validity, yet even these metrics may overestimate real-world attack viability due to simplified laboratory conditions [38].

8.3.2. Transferability analysis

Cross-dataset adversarial transferability demonstrates concerning limitations that undermine claims of attack generalizability [21]. While attacks may achieve high success rates on training datasets, effectiveness drops dramatically when applied to different network environments or NIDS implementations, with transferability rates often below 40% even for sophisticated attacks specifically designed for discrete network domains. The transferability problem becomes particularly acute when attacks trained on specific datasets encounter different feature extraction pipelines, traffic characteristics, or network topologies. Research reveals that transferability failures often correlate with fundamental differences in traffic distributions and feature correlations between datasets rather than model-specific vulnerabilities [149]. Same attack classes occupy different regions of feature space across datasets, creating transferability barriers that suggest attacks exploit dataset-specific artifacts rather than fundamental NIDS vulnerabilities. This finding has profound implications for adversarial robustness evaluation, as attacks that demonstrate high transferability may actually be exploiting common dataset limitations rather than genuine security vulnerabilities [142].

9. Lightweight feature extraction for adversarial robustness

Lightweight feature extraction has emerged as a pivotal technique for achieving adversarial robustness while maintaining computational efficiency in resource-constrained NIDS deployments. Table 6 provides a comprehensive comparative analysis of lightweight feature extraction methods across five major categories: Autoencoder-Based Methods (Section 9.1), Bio-Inspired Optimization Methods (Section 9.2), Information-Theoretic Methods (Section 9.3), Traditional Methods (Section 9.4), and Hardware-Optimized Methods (Section 9.5), systematically evaluating their mathematical foundations, datasets, feature reduction capabilities, accuracy, processing time, memory usage, hardware requirements, adversarial robustness, energy efficiency, and real-time capability. The category labels in Table 6 correspond directly to the subsection headings below to facilitate cross-referencing.

Table 6
Lightweight feature extraction comparative analysis.

Method Category	Technique	Mathematical Foundation	Dataset	Original Features	Reduced Features	Acc. (%)	Feature Reduction (%)	Processing Time (μs)	Memory Usage (MB)	Hardware Requirements	Adversarial Robustness	Energy Efficiency	Real-time Capability	Citations
Autoencoder-Based Methods														
Autoencoder-Based	**Standard Autoencoder**	$L_{reconstruction} = \|x - \hat{x}\|_2^2$	NSL-KDD	41	3–14	89.5 ± 0.8	65.9–92.7	45–60	15–25	CPU/GPU	Moderate	Medium	Medium	[153]
	Sparse Denoising Autoencoder	$L_{sparse} = L_{reconstruction} + \lambda\|h\|_1$	NSL-KDD	41	8–12	97.94 ± 0.3	70.7–80.5	55–75	20–35	GPU preferred	Good	Medium	Medium	[154]
	Variational Autoencoder	$L_{VAE} = L_{reconstruction} + D_{KL}(q(z\|x)\|\|p(z))$	CIC-IDS 2017	83	15–25	94.2 ± 0.5	69.9–81.9	65–85	25–40	GPU required	Moderate robustness	Medium	Medium	[155]
	LSTM-Autoencoder	$h_t = \sigma(W_{hh}h_{t-1} + W_{xh}x_t + b_h)$	IoT datasets	Variable	10–20	92.8 ± 0.6	50–75	80–120	35–55	GPU + memory	Temporal robustness	Medium	Medium	[6]
Bio-Inspired Optimization Methods														
Bio-Inspired	**GMGWO + LDA**	$F(x) = \alpha \cdot Accuracy(x) + \beta \cdot (1 - \|x\|/X_{total})$	NSL-KDD	41	6–12	96.08 ± 0.4	70.7–85.4	25–40	8–15	CPU efficient	Good	Very High	Very High	[108]
	SABPIO	Temperature-controlled binary optimization	Multiple	41	6	92.58 (F1)	85.4	20–35	5–12	CPU only	Good (minimal features)	Very High	Very High	[60]
	SCA-SSA Hybrid	Multi-objective: $\min f_1(x), \min f_2(x)$	UNSW-NB15	49	8–15	94.7 ± 0.5	69.4–83.7	30–50	10–18	CPU efficient	Balanced trade-off	High	High	[65]
	Enhanced Whale Optimization	Position updates with LDA integration	KDD-99	Variable	10–18	97.0 ± 0.3	Variable	35–55	12–20	CPU + moderate memory	High diversity protection	High	High	[156]
Information-Theoretic Methods														
Information-Theoretic	**FSNID Method**	$I(X;Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$	Multiple	Variable	Optimal subset	94.15–95.9	Linear complexity	15–25	8–15	CPU efficient	Principled feature selection	Very High	Very High	[157]
	Mutual Information	$NMI(X,Y) = \frac{I(X;Y)}{\sqrt{H(X)H(Y)}}$	CIC-IDS 2017	83	12–25	93.8 ± 0.7	69.9–85.5	20–35	10–18	CPU only	Nonlinear relationship preservation	Very High	High	[158]
	Conditional MI	$I(X;Y\|Z) = \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y\|z)}{p(x\|z)p(y\|z)}$	Multiple	Variable	Context-dependent	95.2 ± 0.4	Variable	25–40	12–22	CPU + memory	Context-aware	High	High	[159]
Traditional Methods														

9.1. Autoencoder-based methods

Autoencoder-based feature extraction has emerged as a pivotal technique for achieving adversarial robustness while maintaining computational efficiency in resource-constrained NIDS deployments [106]. These approaches leverage the inherent denoising properties of autoencoder architectures to create feature representations that exhibit natural resistance to adversarial perturbations while reducing dimensionality to enable real-time processing in high-throughput network environments [81].

9.1.1. Dimensionality reduction benefits

Autoencoder-based dimensionality reduction demonstrates superior performance compared to traditional Principal Component Analysis (PCA) approaches, achieving approximately 15% improvement in adversarial contamination scenarios while maintain-

Table 6
Cont.

Method Category	Technique	Mathematical Foundation	Dataset	Original Features	Reduced Features	Acc. (%)	Feature Reduction (%)	Processing Time (μs)	Memory Usage (MB)	Hardware Requirements	Adversarial Robustness	Energy Efficiency	Real-time Capability	Citations
	PCA	Eigenvalue decomposition: $C = V\Lambda V^T$	NSL-KDD	41	8–15	87.3 ± 0.9	63.4–80.5	12–20	5–10	CPU only	Poor (linear assumptions)	Very High	Very High	[106]
Traditional	**Feature Selection (Filter)**	Statistical correlation and chi-square	UNSW-NB15	49	10–20	89.4 ± 0.6	59.2–79.6	8–15	3–8	CPU minimal	Limited (static selection)	Very High	Very High	[160]
	Wrapper Methods	Recursive feature elimination	CIC-IDS2017	83	15–30	91.2 ± 0.5	63.9–81.9	40–80	8–15	CPU moderate	Model-dependent	High	High	[161]
Hardware-Optimized Methods														
	Quantized Autoencoder	8-bit/16-bit quantization	Edge devices	Variable	Compressed	96.8 ± 0.4	Model compression	8–15	2–5	Edge TPU/ARM	Quantization-induced robustness	Extremely High	Very High	[162]
Hardware-Optimized	**FPGA Implementation**	Hard- ware pipeline processing	NetFPGA	Variable	Hardware-specific	98.22 ± 0.2	Fixed architecture	5–12	Hardware-fixed	FPGA required	Deterministic processing	High (hardware)	Extremely High	[107]
	Neuromorphic Approach	Spike-based: $\tau_m \frac{dV}{dt} = -(V - V_{rest}) + R_m I(t)$	Theoretical	Variable	Event-driven	Under development	N/A	Event-dependent	Minimal	Neuromorphic chips	Temporal spike robustness	Extremely High	High (event-driven)	[163]

ing interpretability and computational efficiency [106]. Carefully architected autoencoders with latent dimensions ranging from 3–14 achieve F1-scores up to 0.895 on NSL-KDD datasets while providing inherent adversarial denoising capabilities through reconstruction error analysis. The mathematical foundation involves minimizing reconstruction loss: $L_{reconstruction} = ||x - \hat{x}||_2^2$, where $\hat{x} = \text{Decoder}(\text{Encoder}(x))$, combined with regularization terms that encourage robust feature learning. Empirical evaluation reveals that autoencoder-based approaches maintain equivalent accuracy under adversarial conditions while traditional methods degrade up to 50% in F1-score performance. The dimensionality reduction benefits extend beyond computational efficiency to include natural adversarial filtering, as perturbations that do not lie on the learned data manifold are automatically attenuated during the encoding-decoding process [80].

9.1.2. Sparse representation methods

Sparse representation methods employ L_1 sparsity regularization to create feature representations with inherent adversarial robustness through orthogonal constraints that prevent feature correlation exploitation [164]. Sparse deep denoising autoencoders achieve remarkable performance with 97.94% accuracy on NSL-KDD datasets while providing natural adversarial defense through feature diversity mechanisms and reconstruction-based denoising capabilities. The mathematical formulation incorporates sparsity penalties: $L_{sparse} = L_{reconstruction} + \lambda||h||_1$, where h represents the hidden layer activations and λ controls sparsity level. Orthogonal constraints prevent adversarial attacks from exploiting feature correlations, a common vulnerability in traditional dense representations. The sparse representation approach demonstrates particular effectiveness against gradient-based attacks that rely on feature interdependencies for perturbation generation. However, implementation challenges include hyperparameter sensitivity for sparsity regularization and computational overhead during training phases, while the optimal sparsity level varies significantly across different network environments and attack scenarios [165].

9.2. Bio-inspired optimization methods

Bio-inspired optimization algorithms provide sophisticated approaches to feature selection and extraction that leverage evolutionary principles and swarm intelligence to identify optimal feature subsets for adversarial-robust NIDS [156]. These methodologies address the computational complexity of exhaustive feature space exploration while maintaining interpretability and adaptability to diverse network environments through nature-inspired search strategies that balance exploration and exploitation in high-dimensional optimization landscapes [166].

9.2.1. Nature-inspired feature selection

Nature-inspired algorithms demonstrate remarkable effectiveness in identifying optimal feature subsets that balance detection accuracy with adversarial robustness. The Enhanced Whale Optimization Algorithm combined with Linear Discriminant Analysis

achieves 96.08% accuracy on NSL-KDD datasets while dramatically reducing computational complexity through intelligent search space navigation [156]. The Simulated Annealing Binary Pigeon-Inspired Optimizer (SABPIO) represents another breakthrough approach, reducing feature spaces from 41 to 6 dimensions while maintaining 92.58% F1-scores through temperature-controlled probabilistic search mechanisms. The mathematical formulation involves position updates: $X_{i+1} = X_i - A \cdot |C \cdot X_{best} - X_i|$, where A and C represent coefficient vectors that balance global and local search capabilities. The Generalized Mean Grey Wolf Optimization (GMGWO) algorithm incorporates enhanced search pattern flexibility through fitness functions: $F(x) = \alpha \cdot Accuracy(x) + \beta \cdot \left(1 - |x|/|X_{total}|\right)$, that explicitly balance detection performance with feature reduction objectives. These approaches demonstrate particular effectiveness in edge computing architectures where computational resources are severely limited, enabling real-time intrusion detection with minimal hardware requirements [167].

9.2.2. Multi-objective optimization

Multi-objective optimization frameworks address the inherent trade-offs between detection accuracy, computational efficiency, and adversarial robustness through sophisticated evolutionary algorithms that maintain Pareto-optimal solution sets [65]. The hybrid Sine-Cosine Algorithm with Salp Swarm Algorithm (SCA-SSA) approach balances feature reduction with detection accuracy through multi-objective optimization that simultaneously optimizes: $\min f_1(x) = 1 - Accuracy(x)$ and $\min f_2(x) = |Selected_Features|/|Total_Features|$. The Quantum Vaccinated Immune Clonal Algorithm (QVICA) incorporates quantum computing principles with Estimation of Distribution Algorithm for discrimination rule extraction, achieving superior performance across multiple evaluation criteria. Advanced implementations utilize non-dominated sorting optimization that maintains diverse solution populations while converging toward optimal trade-off regions. However, computational complexity scales exponentially with the number of objectives, while solution interpretation becomes challenging when more than three objectives are simultaneously optimized, limiting practical applicability in real-time network security scenarios [61].

9.3. Information-theoretic methods

Information-theoretic approaches provide principled methodologies for feature selection and extraction that leverage mathematical foundations from information theory to identify features with maximum discriminative power while maintaining computational tractability [168]. These approaches overcome limitations of correlation-based methods through rigorous mathematical frameworks that quantify feature relevance, redundancy, and conditional independence relationships essential for adversarial robustness in network intrusion detection [169].

9.3.1. FSNID method

The Feature Selection for Network Intrusion Detection (FSNID) method represents a breakthrough in principled feature relevance assessment, achieving linear time complexity while providing 4.15–5.9% average detection rate improvements over individual feature selection approaches [157]. The mathematical foundation employs neural network function approximation combined with novel information-theoretic measures that overcome traditional correlation limitations through mutual information estimation: $I(X;Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$, where X represents feature variables and Y denotes class labels. The FSNID framework incorporates conditional mutual information analysis to identify feature interactions: $I(X;Y|Z) = \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y|z)}{p(x|z)p(y|z)}$, enabling detection of synergistic feature relationships that individual relevance measures cannot capture. Advanced implementations utilize differential entropy estimation for continuous features combined with discrete entropy calculations for categorical network attributes, providing comprehensive coverage across diverse feature types. The approach demonstrates particular effectiveness in high-dimensional network feature spaces where traditional statistical methods fail due to curse of dimensionality effects, maintaining computational efficiency through approximation algorithms that scale linearly with dataset size [161].

9.3.2. Mutual information analysis

Mutual information analysis provides rigorous mathematical frameworks for quantifying statistical dependencies between network features and attack classifications while identifying redundant feature relationships that can be eliminated without performance degradation [158]. The approach employs entropy-based measures: $H(X) = -\sum_x p(x) \log p(x)$, combined with joint entropy calculations: $H(X,Y) = -\sum_{x,y} p(x,y) \log p(x,y)$ to derive mutual information estimates that capture non-linear feature relationships invisible to correlation analysis. Advanced implementations incorporate normalized mutual information: $\mathrm{NMI}(X,Y) = \frac{I(X;Y)}{\sqrt{H(X)H(Y)}}$, which enables fair comparison across features with different value ranges and distributions. The mathematical framework extends to conditional mutual information networks that model complex feature interdependencies through graphical models, enabling identification of minimal feature sets that preserve classification performance while maximizing adversarial robustness. However, mutual information estimation faces computational challenges in high-dimensional spaces where probability density estimation becomes unreliable, while discrete binning approaches for continuous features may lose important distributional information that affects adversarial vulnerability assessment [170].

9.3.3. Conditional mutual information methods

Conditional Mutual Information (CMI) method lies in its sophisticated ability to quantify the unique information a feature provides relative to a target class, effectively filtering out redundant variables that simpler correlation metrics might overlook [159]. This ensures a highly discriminative yet compact feature subset, which is vital for maintaining detection accuracy in high-dimensional

datasets like UNSW-NB15 or CICIDS2017. However, a notable weakness is its high computational complexity; calculating joint probabilities across multiple variables is resource-intensive, particularly as the feature space expands. Furthermore, a critical limitation is the "data sparsity" problem, where the reliability of CMI estimates degrades significantly when using limited samples or continuous data that requires binning, potentially leading to suboptimal feature selection in dynamic, real-time streaming environments.

9.4. Traditional methods

Traditional feature extraction and selection methods provide established computational baselines for adversarial-robust NIDS, characterised by low resource requirements and high real-time viability despite weaker adversarial robustness guarantees compared to learned representations [6,60]. Table 6 documents three representative approaches spanning dimensionality reduction, filter-based selection, and wrapper-based selection paradigms.

9.4.1. Principal component analysis (PCA)

PCA provides linear dimensionality reduction through eigenvalue decomposition of the feature covariance matrix: $C = V\Lambda V^T$, projecting 41 NSL-KDD features to 8–15 principal components while retaining maximum variance [40]. Within-dataset accuracy reaches 80.5% ± 0.9 with processing times of 5–10 s, and PCA requires CPU-only hardware with minimal memory (15–25 MB), yielding Very High energy efficiency and real-time capability. The adversarial robustness limitation is fundamental: PCA linear projections preserve the adversarial perturbation direction if it aligns with high-variance principal components, and gradient-based attacks can reconstruct the projection basis to generate perturbations that survive dimensionality reduction intact. This Poor adversarial robustness characterisation in Table 6 reflects the method's inability to attenuate adversarial directions that coincide with legitimate traffic variance.

9.4.2. Filter-based feature selection

Filter-based methods rank and select features by statistical relevance criteria (correlation, chi-square, information gain) independently of the downstream classifier, reducing 49 UNSW-NB15 features to an optimal subset [108]. This classifier-independence produces Very High real-time capability and minimal CPU requirements, with within-dataset accuracy of 89.4% ± 0.6 at 8–15 s processing time. Adversarial robustness is Limited because static feature selection is computed once from benign training data: an adversary who knows the selected feature subset can concentrate perturbations on those features specifically, and the selection does not adapt to adversarial distribution shifts. The approach is appropriate for resource-constrained deployments where computational budget prohibits learned representations, with the understanding that adversarial robustness must be supplemented by other defense layers.

9.4.3. Wrapper methods

Wrapper methods evaluate feature subsets by classifier performance, using recursive feature elimination or evolutionary search to identify subsets that maximise the specific classifier's detection accuracy [60,65]. Applied to 83 CIC-IDS2017 features, wrapper methods achieve 91.2% ± 0.5 accuracy with 8–15 s processing at moderate CPU memory requirements. Adversarial robustness is characterised as Model-Dependent in Table 6: because wrapper selection optimises for classifier performance on clean data, the resulting subset is no more adversarially robust than the underlying classifier. However, wrapper methods can be extended to adversarial wrapper selection by including adversarial examples in the evaluation loop, converting the static selection into an adaptive defense mechanism. The primary practical limitation is computational cost: wrapper evaluation requires re-training or re-evaluating the classifier for each candidate feature subset, which becomes prohibitive for high-dimensional spaces without heuristic search strategies.

9.5. Hardware-optimized methods

Hardware implementation considerations become critical for deploying adversarial-robust NIDS in production environments where real-time processing constraints, power limitations, and cost factors fundamentally shape the feasibility of sophisticated defense mechanisms [107]. These considerations bridge the gap between theoretical adversarial robustness and practical deployment capabilities, revealing how hardware constraints influence the effectiveness of lightweight feature extraction approaches in real-world network security scenarios [171].

9.5.1. FPGA deployments

Field-Programmable Gate Array (FPGA) implementations demonstrate production-ready capabilities for adversarial-robust NIDS with remarkable performance characteristics [107]. NetFPGA-SUME platforms achieve throughput exceeding 10 Gbps while maintaining 98.22% accuracy on CIC-IDS2017 datasets, utilizing only 20–21% Look-Up Table (LUT) usage at operating frequencies of 200–206 MHz with power consumption ranging from 10.32-11.27W [172]. The parallel processing elements enable direct packet-to-hardware processing that eliminates software bottlenecks plaguing traditional CPU-based implementations. Mathematical operations are implemented through dedicated hardware multipliers and block RAM structures that provide deterministic processing latencies essential for real-time network security applications. Advanced FPGA implementations incorporate pipelined architectures where feature extraction, classification, and output generation occur simultaneously across different pipeline stages, enabling continuous packet processing without buffering delays. However, hardware resource constraints limit the complexity of deployable algorithms,

while reconfiguration overhead for adaptive defense mechanisms requires careful balance between flexibility and performance. The fixed-point arithmetic requirements introduce quantization errors that may affect adversarial robustness, necessitating specialized number representation schemes that maintain detection accuracy while fitting within hardware constraints [173].

9.5.2. Edge computing optimization

Edge computing implementations showcase remarkable efficiency gains for adversarial-robust NIDS through specialized hardware acceleration and optimized algorithm deployment [162]. NVIDIA Jetson Nano deployments achieve 99.43% accuracy with processing times optimized for edge deployment scenarios, while quantized QMLP models provide automotive CAN intrusion detection with minimal resource utilization suitable for embedded systems. The optimization strategies include quantized neural networks that provide 1.9x training acceleration while maintaining or improving adversarial robustness compared to full-precision models through quantization-induced regularization effects. Memory-efficient implementations utilize model compression techniques including pruning, quantization, and knowledge distillation that reduce model size by 8-32x ratios required for practical edge deployment. However, edge computing faces fundamental trade-offs between model complexity and available computational resources, while communication constraints limit the sophistication of federated learning approaches that could enhance adversarial robustness through collaborative intelligence. Battery-powered IoT devices impose additional energy constraints that restrict the feasible complexity of adversarial defense mechanisms, requiring careful algorithm selection that balances security effectiveness against power consumption requirements [174].

10. Future research directions

10.1. Quantum-resistant approaches

The emergence of quantum computing represents a fundamental threat to current cryptographic foundations underpinning network security, necessitating quantum-resistant feature engineering and adversarial defense mechanisms for NIDS. Post-quantum cryptographic considerations become increasingly critical as quantum algorithms like Shor's algorithm threaten RSA and elliptic curve cryptosystems that secure network communications, while Grover's algorithm reduces the effective security of symmetric cryptographic primitives by half [175]. Quantum-resistant adversarial techniques must address both the computational advantages that quantum adversaries may possess and the new attack vectors that quantum-classical hybrid systems introduce. Research into quantum machine learning algorithms reveals potential vulnerabilities where quantum speedups could enable more efficient adversarial example generation, necessitating defensive mechanisms that account for quantum computational capabilities. However, quantum-resistant NIDS development faces significant challenges including the computational overhead of post-quantum cryptographic primitives, uncertainty regarding quantum hardware capabilities and timelines, and the fundamental difficulty of designing defenses against theoretical quantum attack capabilities that remain largely speculative.

10.2. Neuromorphic computing integration

Neuromorphic computing architectures offer potential for ultra-low power adversarial defense mechanisms in IoT deployments through brain-inspired processing paradigms that fundamentally differ from traditional von Neumann architectures. These systems employ event-driven processing and spike-based neural networks that consume power only when processing occurs, achieving energy efficiency improvements of 1000× compared to conventional digital processors for specific computational tasks [176]. The mathematical foundation involves leaky integrate-and-fire neuron models: $\tau_m \frac{dV}{dt} = -(V - V_{rest}) + R_m I(t)$, where membrane potential V integrates input currents over time, generating spikes when threshold conditions are met. Neuromorphic implementations demonstrate particular promise for temporal pattern recognition in network traffic analysis, where spike-timing-dependent plasticity enables adaptive learning of attack signatures without explicit retraining procedures. However, neuromorphic computing remains largely theoretical for NIDS applications, with limited hardware availability, immature software frameworks, and unclear scalability properties for high-throughput network processing. The analog nature of neuromorphic computation introduces noise sensitivity issues that may compromise adversarial robustness.

10.3. Federated robustness

Distributed NIDS deployments require sophisticated evaluation frameworks addressing cross-domain transferability of defense mechanisms, privacy-preserving adversarial training, and collaborative threat intelligence sharing across organizational boundaries. The fundamental challenge lies in developing adversarial robustness that generalizes across heterogeneous network environments while preserving data privacy and organizational autonomy. Cross-domain transferability remains poorly understood, with defense mechanisms achieving high robustness within single organizational contexts often failing when deployed across diverse network topologies and traffic patterns. Privacy-preserving adversarial training techniques must balance collaborative learning benefits against information leakage risks, requiring sophisticated cryptographic protocols including secure multi-party computation and homomorphic encryption that introduce substantial computational overhead. Collaborative threat intelligence sharing faces challenges in standardizing adversarial attack representations across different NIDS implementations while maintaining competitive advantages. The mathematical framework involves differential privacy mechanisms: ϵ-differential privacy ensures $\Pr[M(D) \in S] \leq e^{\epsilon} \Pr[M(D') \in S]$

for neighboring datasets D and D', though determining optimal privacy budgets remains challenging. Current federated learning approaches demonstrate average F1-scores of 0.97 across distributed networks, yet robustness evaluation under adversarial conditions reveals significant vulnerabilities when attackers coordinate across multiple participating organizations [118].

10.4. Automated defense evolution

The rapid pace of threat evolution demands automated defensive adaptation through meta-learning frameworks that enable immediate response to emerging attack patterns without human intervention. Traditional manual security updates prove inadequate when adversaries deploy adaptive strategies that evolve faster than human response capabilities, necessitating autonomous defense systems capable of self-modification and continuous improvement. Meta-learning approaches for rapid threat adaptation enable systems to learn optimal learning strategies rather than fixed defense mechanisms, achieving 85% detection rates for zero-day attacks through few-shot learning paradigms [134]. Continuous concept drift detection algorithms automatically identify environmental changes indicating new threat landscapes: statistical tests including Kolmogorov-Smirnov and Mann-Whitney U detect distribution shifts with significance levels $\alpha = 0.05$, triggering adaptive responses when p-values fall below thresholds [177]. Self-evolving defensive architectures employ genetic programming to automatically generate and optimize defense strategies, though implementation remains challenging due to computational complexity and safety concerns regarding autonomous security modifications. The mathematical framework involves online learning with regret bounds: $\sum_{t=1}^{T} \ell_t(w_t) - \min_w \sum_{t=1}^{T} \ell_t(w) \leq O(\sqrt{T})$ ensuring performance guarantees. However, automated evolution faces critical challenges including verification of evolved defenses and preventing adversarial manipulation of the evolution process itself.

11. Critical analysis and research gaps

11.1. Fundamental limitations

Contemporary adversarial NIDS research exhibits systematic evaluation deficiencies that overstate attack efficacy and understate defensive capability [152]. Laboratory evaluation environments fail to capture production network complexity: controlled experimental settings ignore dynamic traffic patterns, infrastructure heterogeneity, and real-time processing constraints that govern operational deployment [123]. Evaluation methodologies predominantly assess attack success in feature space without validating packet-level protocol compliance or semantic preservation – empirical analysis reveals that up to 80.3% of adversarial examples become invalid when domain-specific constraints are properly enforced [11]. Scalability constraints compound these issues, as sophisticated defense mechanisms require 2–10× additional computational resources compared to baseline systems [123], with ensemble methods reaching up to 10× overhead [15], fundamentally precluding deployment in production environments processing millions of flows per second. Cross-dataset evaluation is rarely performed despite evidence of catastrophic generalization failures [21], while statistical significance testing remains inconsistent and reproducibility of claimed results severely limited [152].

11.2. Theoretical vs. practical gaps

The translation of adversarial attacks from feature space to valid network packets reveals fundamental limitations that dramatically constrain practical threat viability [11]. Protocol compliance requirements, TCP/UDP header interdependencies, checksum validation, and state machine constraints eliminate the majority of theoretical perturbation possibilities: 60–70% of desired feature modifications lack valid packet representations, and infrastructure filtering mechanisms automatically reject 60–80% of constraint-violating traffic [11]. Semantic preservation failures further reduce the practical attack space, as adversarial modifications that satisfy protocol constraints frequently neutralize the malicious payload during evasion, rendering attacks functionally ineffective in operational environments. These findings suggest that published adversarial attack success rates, typically measured in feature space under unconstrained conditions, substantially overestimate real-world vulnerability [11]. Conversely, defense deployment faces integration complexity, maintenance overhead, and real-time constraint violations that create equivalent barriers between theoretical defensive capability and practical security enhancement [123].

11.3. Reproducibility crisis

The adversarial NIDS field faces a reproducibility crisis of particular severity, with systematic evidence that up to 80% of adversarial robustness claims cannot be replicated under rigorous independent replication [152]. Software incompatibilities across machine learning frameworks, version conflicts, platform-specific implementations, and documented bugs in foundational tools such as CICFlowMeter – which underlies the majority of benchmark datasets – prevent consistent experimental reproduction [139,152]. Evaluation protocol inconsistencies compound these issues: attack success rates are calculated under different constraint enforcement levels, with differing preprocessing procedures and metric definitions that produce incomparable results even across studies ostensibly evaluating identical phenomena [123]. Ground truth verification lacks standardization, with no consensus regarding valid attack data characteristics or constraint compliance criteria. Resolving this crisis requires mandatory artifact sharing (complete repositories, preprocessed datasets, dependency specifications), standardised evaluation metrics with precisely defined constraint enforcement protocols, and cross-dataset validation as the minimum acceptable evaluation standard [21,152].

Table 7
Research gaps, future directions, and practical recommendations for adversarial NIDS. Synthesised from Section 10 (future research), Section 11 (critical analysis), and Section 12 (practical recommendations). Section references in brackets indicate evidence source within this survey.

Category	Research Gap (Section 11)	Root Cause	Future Research Direction (Section 10)	Recommendation (Section 12
Evaluation Methodology	Feature-space attack success rates overstate real-world viability by treating protocol-invalid perturbations as valid adversarial examples	Evaluation pipelines decouple feature manipulation from packet-level validity; 80.3% of adversarial examples become invalid under domain constraints Section 11.1	Constraint-enforcing evaluation benchmarks that report packet-valid attack success rates separately from feature-space success rates	Mandate dual-metric reporting: feature-space evasion rate *and* packet-valid evasion rate; reject papers that conflate the two
	Static threat models misrepresent adaptive adversaries who modify attack strategies in response to defensive actions	Most evaluations test oblivious adversaries only; no standardised tiered adversary framework exists across the field Section 11.1	Three-tier adversary evaluation protocol: oblivious, informed, and iteratively adaptive—reported separately per defense	Establish an adversary-tier reporting standard analogous to the Common Vulnerability Scoring System; conferences should require all three tiers
	Cross-dataset generalisation failures are systematically hidden by single-dataset evaluation; 30–40% accuracy under cross-dataset conditions vs. 95%+ within-dataset	Benchmark datasets (CIC-IDS2017, UNSW-NB15) have distribution characteristics too narrow to represent diverse operational traffic Section 8.1	Multi-domain evaluation frameworks combining heterogeneous datasets including 5G-NIDD, CICIoMT2024, and enterprise traffic captures	Require cross-dataset validation as a condition of publication; single-dataset results reportable only as preliminary evidence
Attack Translation	Inverse feature-mapping problem: 60–70% of desired feature perturbations have no valid packet-level representation, fundamentally constraining practical attack space	Attack generation operates in continuous feature space; network packets occupy a discrete, protocol-constrained manifold that gradient methods cannot navigate directly Section 5.3	Problem-space attack generators that use protocol grammars and semantic constraint solvers as the primary optimisation domain, not feature space	Fund research into differentiable protocol simulators that can back-propagate attack gradients through packet construction constraints
	Semantic preservation failure: adversarial modifications frequently neutralise malicious functionality during evasion, producing non-functional attacks	Feature perturbation objectives optimise statistical distance from classifier decision boundary without modelling payload execution semantics Section 5.4	Semantics-aware attack frameworks that jointly optimise for evasion and functional attack preservation using execution sandbox feedback	Require semantic functionality verification as part of attack evaluation; reported evasion rates must exclude non-functional adversarial examples
defense Deployment	Scalability cliff: advanced defenses (ensemble, certified) require 2–10× computational overhead and 150–500 ms latency, categorically excluding them from sub-millisecond production environments	defense optimisation objectives maximise adversarial accuracy without latency or overhead constraints; deployment feasibility treated as a post-publication concern Section 11.1	Constrained defense optimisation: maximise adversarial robustness subject to ≤20% overhead ceiling and ≤1 ms latency bound as hard constraints	Introduce deployability ceiling as a mandatory evaluation criterion; defenses exceeding thresholds categorised as research-stage, not deployment-ready

Table 7
Cont.

Category	Research Gap (Section 11)	Root Cause	Future Research Direction (Section 10)	Recommendation (Section 12
	Infrastructure dependency barrier: MTD and federated defenses presuppose SDN hardware and cross-organisational coordination absent from 95% of operational environments	defense designs target theoretical best-case infrastructure rather than commodity deployments; no incremental adoption pathway is specified Section 11.2	Layered defense architectures that deliver meaningful robustness improvement within commodity hardware, with optional enhancement tiers for SDN-capable environments	Mandate deployment pathway analysis in every defense paper: minimum infrastructure requirements and the percentage of operational environments that satisfy them
	Brittle degradation: ensemble and detection-based defenses collapse catastrophically (>15% accuracy drop) when any component is compromised or faces adaptive adversaries	defenses are evaluated against fixed, oblivious attack configurations; adaptive and component-failure modes are not part of standard evaluation Sections 6.6 and 11.2	Graceful-degradation-by-design: defense architectures with provable performance floors under partial component failure and adaptive adversarial pressure	Report degraded-mode performance (50% component availability, adaptive adversary) alongside peak performance; brittle defenses should not be presented as robust
Reproducibility	Up to 80% of adversarial robustness claims fail rigorous replication due to undocumented hyperparameter sensitivity, tool dependencies, and constraint enforcement variation	No standardised reproducibility protocol exists for adversarial NIDS; constraint enforcement level—the primary driver of performance variation—is rarely specified Section 11.3	Standardised Adversarial NIDS Reproducibility Protocol specifying: constraint enforcement tier, dataset preprocessing pipeline, statistical testing requirements, and artefact sharing format	Adopt a reproducibility checklist modelled on NeurIPS/ICML standards but extended with network-domain constraint enforcement specifications as a submission requirement
	Metric incomparability: attack success rates calculated under different constraint enforcement levels produce incomparable results across studies, preventing cumulative scientific progress	No consensus definition of "valid adversarial example" in network security; each study applies its own implicit constraint threshold Section 11.3	Shared adversarial validity taxonomy with four tiers: feature-space only, protocol-valid, semantics-preserving, and infrastructure-surviving—reported as separate metrics	Journals and conferences adopt adversarial validity tier as a required reporting field; meta-analyses possible only within the same tier
Emerging Threat Surfaces	5G network slicing, IoT heterogeneity, and IIoT protocol diversity are underrepresented in current adversarial NIDS research; existing defenses are not validated in these environments	Benchmark datasets (CIC-IDS2017, UNSW-NB15) capture Ethernet/IP environments; 5G-NIDD, CICIoMT2024, TON-IoT represent emerging surfaces with distinct constraint profiles Section 8.1	Domain-specific adversarial NIDS frameworks for 5G core networks, healthcare IoMT, and industrial IIoT with constraint profiles matched to each domain's protocol stack	Allocate dedicated evaluation tracks at venues to domain-specific adversarial NIDS; require cross-domain validation for papers claiming generalised robustness
	Foundation-model-assisted adversarial generation (LLM-crafted payloads, diffusion-based traffic synthesis) is not addressed by any current defense category	Existing defenses were designed against gradient-based and GAN-based attacks; LLM-generated attacks can satisfy semantic and protocol constraints natively, bypassing the 80.3% invalidity barrier Section 10.4	defense architectures co-designed with LLM-aware threat models; behavioural analysis beyond statistical features to detect semantically coherent but malicious traffic patterns	Initiate a dedicated research programme on LLM-assisted adversarial NIDS; existing evaluation benchmarks are insufficient and require fundamental extension

11.4. Operationally realistic attack models and evaluation protocols

The four preceding research directions address the defensive landscape of adversarial NIDS. An equally important frontier – largely neglected in current literature – concerns the attack side: developing adversarial attack models and evaluation protocols that reflect the operational realities faced by real network adversaries rather than the idealised laboratory conditions that dominate current research. The systematic analysis presented in this survey demonstrates that laboratory-evaluated attacks overstate real-world threat severity by factors of three to twenty [11,12], making the calibration of both attack design and evaluation methodology a critical research priority.

11.4.1. Operationally grounded attack models

The dominant paradigm in adversarial NIDS research — white-box gradient-based attacks on continuous feature representations — reflects laboratory convenience rather than realistic adversary capabilities [171]. Future attack model design should be governed by three operational constraints that real attackers face but that current research ignores.

First, realistic adversaries operate under query budget constraints: they cannot query a deployed NIDS arbitrarily without triggering rate-limiting or anomaly detection mechanisms. Future black-box attack models should incorporate explicit query-efficiency objectives, building on the finite difference estimation framework $\hat{\nabla} f(x) \approx \frac{f(x+\delta e_i)-f(x-\delta e_i)}{2\delta}$ [46], but extending it with adaptive query scheduling that minimises detection risk while maximising gradient approximation accuracy.

Second, future attack models must be designed for discrete and categorical feature spaces, as boolean flags, protocol numbers, and port identifiers comprise 43–52% of typical NIDS feature sets [12]. Evolutionary and swarm-intelligence methods — including Genetic Algorithms (87–95% lab success on NSL-KDD) and Particle Swarm Optimisation (80–90% on CIC-IDS2017, Table 2) — offer natural discrete-space optimisation and deserve development as primary attack paradigms rather than secondary alternatives to gradient methods.

Third, the adaptive attack paradigm described in Section 7 should become the standard threat model rather than an advanced variant. Real adversaries observe defensive responses and iteratively refine strategies [122]; attack models that assume a static, single-shot perturbation fundamentally misrepresent operational threats. Future work should develop standardised adaptive attack benchmarks – analogous to the AUTOATTACK benchmark in image adversarial learning – that evaluate NIDS defenses against adversaries with adaptive feedback rather than fixed perturbation budgets.

11.4.2. Constraint-aware evaluation protocols

The finding that up to 80.3% of adversarial examples become invalid under realistic domain constraints [12] exposes a fundamental inadequacy in current single-metric evaluation: reporting feature-space attack success rates without packet-level validation systematically overstates adversarial vulnerability. Future evaluation protocols must adopt a tiered validation framework that escalates through four levels of operational realism.

Tier 1 — Feature-Space Baseline: Measure attack success rate against the ML classifier using perturbed feature vectors. This remains a useful upper bound but must be explicitly labelled as unconstrained and non-deployable [123].

Tier 2 — Packet-Space Validity: Apply the inverse feature-mapping step and report the proportion of adversarial feature vectors that correspond to at least one valid network packet satisfying protocol constraints. This metric directly quantifies the many-to-none mapping infeasibility identified in Section 5.3 and should accompany Tier 1 metrics in all future publications.

Tier 3 — Live Testbed Evaluation: Inject constraint-valid adversarial packets into an emulated or physical network stack (e.g., Mininet, GNS3) and measure end-to-end evasion including infrastructure filtering and firewall inspection. This tier captures the 60–80% of constraint-violating packets automatically rejected by network infrastructure [123].

Tier 4 — Production Shadow Evaluation: Replay adversarial traffic against a shadow-mode NIDS on captured real network traffic, addressing the temporal relevance problem where static datasets fail to capture concept drift.

Standardising this four-tier framework as a community reporting norm — analogous to Table 3 in this survey — would prevent the misreporting of unconstrained feature-space success rates as evidence of real-world NIDS vulnerability.

12. Practical recommendations

12.1. For researchers

Addressing the systematic limitations identified in Section 11 requires methodological reforms centred on three priorities [123]. First, cross-dataset adversarial evaluation must replace single-dataset validation as the minimum credible standard, with temporal validation against future traffic data superseding traditional held-out splits that fail to capture concept drift [21]. Second, constraint enforcement protocols must reflect realistic deployment conditions – protocol compliance, infrastructure filtering, and semantic preservation requirements – so that reported attack success rates correspond to operationally viable threats rather than unconstrained feature-space manipulations [11]. Third, comprehensive artifact sharing including complete code repositories, preprocessed datasets, and explicit dependency specifications should become mandatory to enable result replication and build cumulative scientific progress [152]. Statistical significance testing must incorporate multiple comparison corrections, and confidence intervals should account for cross-dataset variability rather than within-dataset performance, given the severe generalization failures documented in Section 8.

12.2. For practitioners

Security practitioners should calibrate adversarial threat assessments against the constraint compliance evidence documented in Sections 3 and 11: the 80.3% invalidity rate of published adversarial examples under realistic constraints [11] suggests that many theoretical attack vectors may not materialise in operational environments with standard infrastructure filtering and protocol enforcement. Deployment decisions should prioritise protocol-aware feature engineering and lightweight ensemble approaches [15] over computationally intensive certified defenses that exceed real-time processing requirements. Incremental deployment strategies – layering adversarial-robust feature extraction atop existing signature-based detection rather than replacing it – provide risk-proportionate security enhancement without the architectural overhaul that advanced defenses often demand [123]. Risk assessments should incorporate computational budget constraints explicitly, as microsecond-level response requirements in high-throughput environments eliminate most sophisticated defense mechanisms before security evaluation begins [107].

12.3. For organizations

Organisations should recognise adversarial robustness as a long-term strategic investment rather than a discrete procurement decision, given the adaptive nature of adversarial threats and the rapid evolution of both attack and defense capabilities documented throughout this survey. Internal expertise in adversarial machine learning is critical for effective threat assessment and vendor evaluation, as published performance metrics are typically measured under conditions that overstate practical protection [152]. Procurement criteria should require adversarial robustness evaluation under realistic constraint enforcement protocols rather than accepting vendor-reported feature-space attack success rates. Infrastructure investment priorities should favour SDN capabilities and programmable network hardware that enable deployment of adaptive defenses such as Moving Target Defense [93] and federated learning-based robustness [94], whose practical viability is otherwise limited by infrastructure dependencies documented in Section 6. Policy frameworks should mandate regular adversarial robustness re-evaluation as threat landscapes and network environments evolve [123].

12.4. Deployment-oriented design principles for defense researchers

The analysis across Sections 6 and 11, and Table 4 reveals that the field's primary failure is not technical inadequacy but misaligned research objectives: defenses are designed to maximise adversarial accuracy under controlled conditions rather than to function within the operational constraints that govern real NIDS deployments. This subsection distils that finding into four actionable principles that should govern future defense design.

12.4.1. Constraint-co-design rather than constraint-compliance

The 80.3% adversarial example invalidation rate established in Section 5 is this survey's most consequential finding – yet defense research continues to treat protocol and infrastructure constraints as post-hoc validity filters rather than as primary design inputs. defense researchers should invert this logic: begin with the constraint envelope (protocol compliance, semantic preservation, infrastructure filtering, real-time latency) and design the defense mechanism within those bounds from the outset. A defense co-designed with network protocol semantics will, by construction, address the same constraint set that defeats most adversarial attacks – yielding mechanisms that are simultaneously harder to evade and deployable without architectural change.

12.4.2. Replace accuracy maximisation with viability-bounded optimisation

Section 11.1 demonstrates that ensemble methods achieving 2-10 computational overhead and certified defenses introducing 300-500% overhead are not deployment failures – they are optimisation failures, because no deployment constraint was set before the optimisation objective was defined. Future defense research should adopt a constrained optimisation framing: maximise adversarial robustness subject to an explicit overhead ceiling ($\leq$20% above baseline processing cost) and a latency bound ($\leq$1ms for flow-level classification). This reframes the research contribution from 'highest accuracy reported' to 'highest accuracy within operational feasibility' – a more meaningful and reproducible benchmark.

12.4.3. Adversarial adaptivity as a first-class evaluation criterion

Section 11.2 identifies that static threat models fundamentally misrepresent adaptive adversaries, yet defense evaluations overwhelmingly report performance against fixed attack configurations. defense researchers should evaluate mechanisms against three adversary tiers: (a) oblivious adversaries unaware of the defense, (b) informed adversaries with full knowledge of the defense architecture, and (c) adaptive adversaries who iteratively modify attack strategies in response to defense behaviour. defenses that degrade catastrophically under informed or adaptive adversaries – regardless of oblivious-adversary accuracy – should not be presented as robust solutions. Graceful degradation under adaptive pressure is a more reliable deployment predictor than peak accuracy under idealised conditions.

12.4.4. Deployment pathway as a mandatory research output

Section 11.2 identifies legacy system integration and infrastructure replacement as the dominant deployment barriers. For every proposed defense, researchers should specify an explicit deployment pathway: the minimum infrastructure changes required, the percentage of operational environments that satisfy those requirements without modification, and the incremental adoption path from

existing commodity NIDS deployments. defenses that cannot integrate as additive layers within existing architectures should be clearly categorised as long-horizon research rather than near-term solutions – a distinction the field currently obscures by evaluating all mechanisms under the same framework regardless of deployment readiness. Table 7 consolidates all research gaps, future directions, and practical recommendations across Sections 10-12 by priority.

13. Conclusion

Based on the comprehensive literature review examining adversarial attacks against Network Intrusion Detection Systems, this analysis reveals a critical disconnect between theoretical attack capabilities and practical implementation constraints. While gradient-based attacks, generative adversarial approaches, and advanced frameworks demonstrate remarkable evasion effectiveness in laboratory conditions–often exceeding 97% success rates–the translation to real-world network environments faces severe limitations. The systematic analysis demonstrates that up to 80.3% of adversarial examples become invalid when domain-specific constraints including protocol compliance, semantic preservation, and infrastructure filtering are properly enforced. Defense mechanisms including adversarial training, ensemble methods, and moving target defense strategies show promise for enhancing robustness, yet face significant scalability challenges that limit production deployment. The evaluation reveals fundamental limitations in current research methodologies, including dataset quality issues, reproducibility challenges, and evaluation protocols that fail to capture real-world deployment constraints. These findings suggest that many published adversarial attacks substantially overstate real-world vulnerabilities of ML-based NIDS, providing a more nuanced understanding of practical threat landscapes. Several limitations contextualize this survey's findings. First, the scope is restricted to literature published through 2025, potentially excluding rapid, subsequent developments in adversarial machine learning. Second, the focus remains strictly on machine learning-based NIDS and network-layer attacks, leaving rule-based systems and encrypted traffic underrepresented. Third, quantitative conclusions rely on standard benchmark datasets like CIC-IDS2017 and UNSW-NB15, which may not generalize to diverse operational environments. Finally, findings synthesize self-reported results without independent replication, necessitating caution when interpreting cross-study performance comparisons. The research underscores the importance of constraint-aware evaluation methodologies and highlights the need for defense mechanisms specifically designed for network security domains. Future work must prioritize developing realistic evaluation frameworks that bridge the theory-practice gap while addressing the unique challenges posed by network protocol semantics and operational deployment requirements.

CRediT authorship contribution statement

Md Mehedi Hasan: Writing – review & editing, Methodology, Formal analysis, Data curation, Conceptualization; **Rafiqul Islam:** Supervision; **Quazi Mamun:** Writing – review & editing, Supervision; **Md Zahidul Islam:** Writing – review & editing, Supervision; **Junbin Gao:** Writing – review & editing, Supervision.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Md Mehedi Hasan reports financial support was provided by Charles Sturt University. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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